SSD1963

Advance Information

1215KB Embedded Display SRAM
LCD Display Controller

This document contains information on a new product. Specifications and information herein are subject to change without notice.

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Appendix: IC Revision history of SSD1963 Specification

Version	Change Items	Effective Date
0.10 21-Nov-08	1st Release	24-Nov-08
0.10 08-Dec-08	1. Changed the set_pll_mnk to set_pll_mn in section 7.2 2. Change register name in section 8 3. Removed ABC 4. Revised description for REG 0x00, 0x01, 0x0C, 0x0D, 0x0E, 0x10, 0x11, 0x21, 0x26, 0x28, 0x2A, 0x2B, 0x2C, 0x2E, 0x33, 0x34, 0x35, 0x36, 0x37, 0x3A, 0x3C, 0x3E, 0x44, 0x45, 0xA1, 0xB0, 0xB1, 0xB4, 0xB5, 0xB6, 0xB7, 0xB8, 0xB9, 0xBE, 0xBF, 0xD0, 0xD1, 0xD4, 0xE5. 5. Added max VIH in Table 12-1 6. Added Table 9-1 7. Added Table 11-1 8. Revised Figure 9-19 9. Revised Figure 14-2 10. Revised Figure 13-5 11. Corrected typo for Table 7-2 12. Revised test condition for 12 and 13	10-Dec-08
1.0 07-May-09	1. Changed status to Advance Information 2. Update min/max rating of VDDD and VDDPLL in Table 11-1 3. Added tape and reel drawing of 128-pin LQFP package in Section 15.3 4. Revised Section 13.2 5. 5. Added 12 bits for Table 7-1 6. Removed TTL interface 7. Revised section 7.1.5 8. Change the title of section 7.2 9. Revised command description in section 8 10. Removed the command 0x0C and 0x3A 11. Added figures in section 13.4 12. Revised figures in section 13.3 13. Revise Table 6-1	18-May-09
1.1 23-Dec-09	1. Update Table 7-1 2. Revised section 9.72	18-Jan-10

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1 GENERAL DESCRIPTION

SSD1963 is a display controller of 1215K byte frame buffer to support up to 864 x 480 x 24bit graphics content. It also equips parallel MCU interfaces in different bus width to receive graphics data and command from MCU. Its display interface supports common RAM-less LCD driver of color depth up to 24 bit-per-pixel.

2 FEATURES

- Display feature
  - Built-in 1215K bytes frame buffer. Support up to 864 x 480 at 24bpp display
  - Support TFT 18/24-bit generic RGB interface panel
  - Support 8-bit serial RGB interface
  - Hardware rotation of 0, 90, 180, 270 degree
  - Hardware display mirroring
  - Hardware windowing
  - Programmable brightness, contrast and saturation control
  - Dynamic Backlight Control (DBC) via PWM signal
- MCU connectivity
  - 8/9/16/18/24-bit MCU interface
  - Tearing effect signal
- I/O Connectivity
  - 4 GPIO pins
- Built-in clock generator
- Deep sleep mode for power saving
- Core supply power (V_{DDPLL} and V_{DDD}): 1.2V±0.1V
- I/O supply power(V_{DDIO}): 1.65V to 3.6V
- LCD interface supply power (V_{DDLCD}): 1.65V to 3.6V

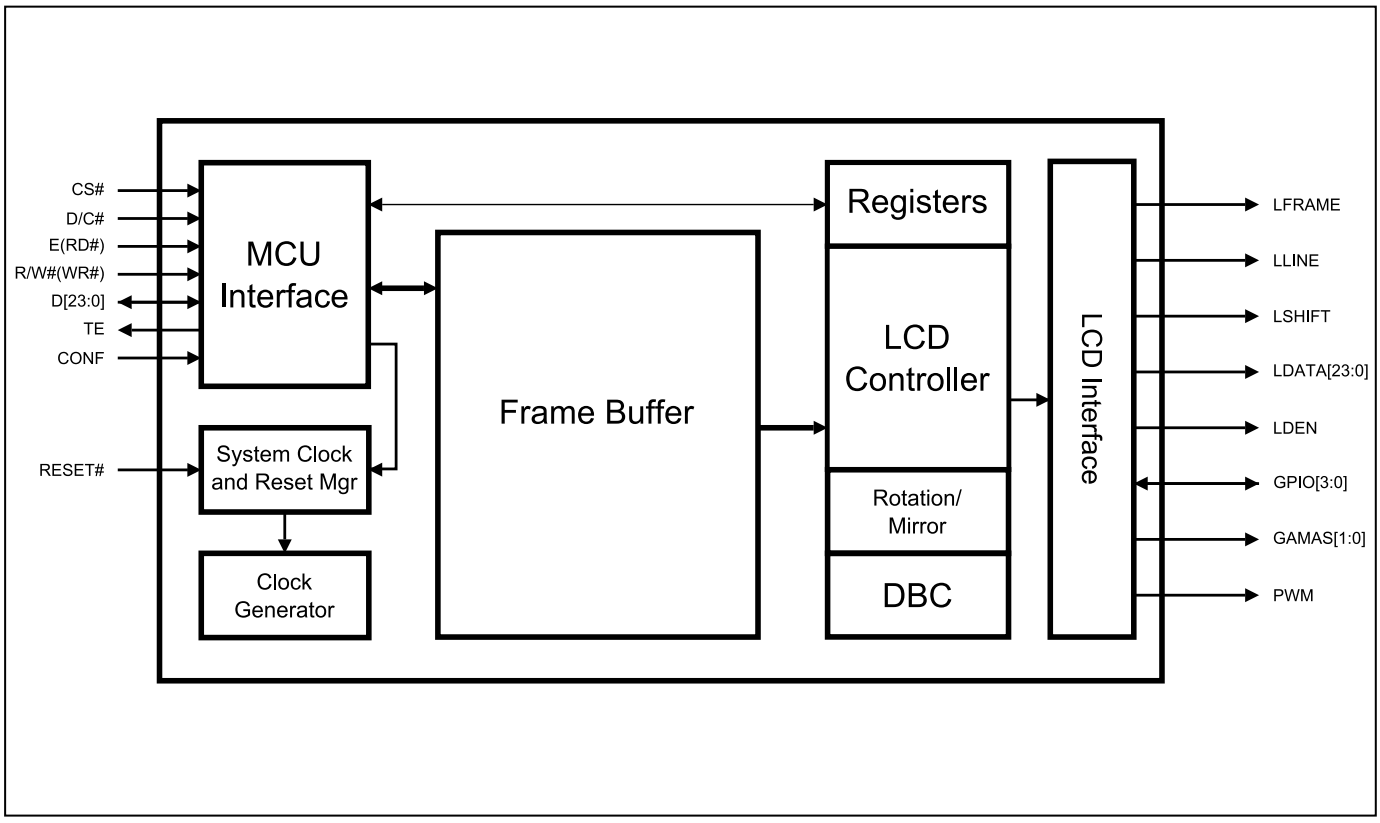
3 ORDERING INFORMATION

Table 3-1: Ordering Information

Ordering Part Number	Package Form
SSD1963G41	TFBGA-80 (Tray)
SSD1963QL9	LQFP-128 (Tray)
SSD1963QL9R	LQFP-128 (Tape & Reel)

4 BLOCK DIAGRAM

Figure 4-1: SSD1963 Block Diagram



5 PIN ARRANGEMENT

5.1 80 balls TFBGA

Figure 5-1: Pinout Diagram –TFBGA (Topview)

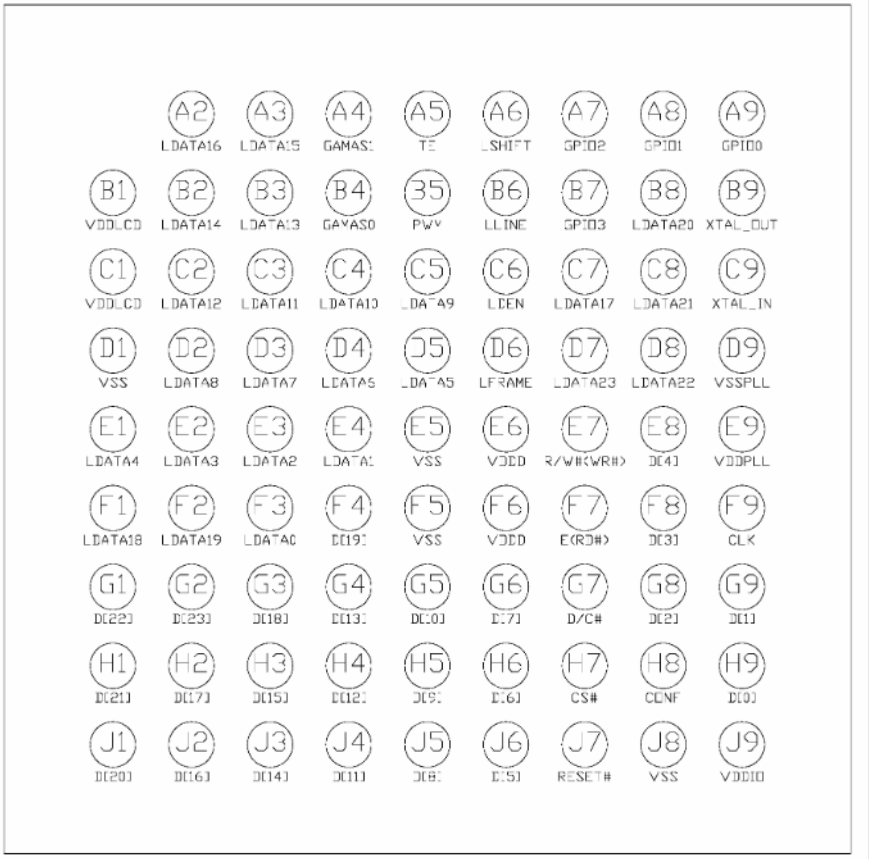


Table 5-1: TFBGA Pin Assignment Table

Pin #	Signal Name	Pin #	Signal Name	Pin #	Signal Name	Pin #	Signal Name	Pin #	Signal Name
A1	-	C1	VDDLCD	E1	LDATA4	G1	D[22]	J1	D[20]
A2	LDATA16	C2	LDATA12	E2	LDATA3	G2	D[23]	J2	D[16]
A3	LDATA15	C3	LDATA11	E3	LDATA2	G3	D[18]	J3	D[14]
A4	GAMAS1	C4	LDATA10	E4	LDATA1	G4	D[13]	J4	D[11]
A5	TE	C5	LDATA9	E5	VSS	G5	D[10]	J5	D[8]
A6	LSHIFT	C6	LDEN	E6	VDDD	G6	D[7]	J6	D[5]
A7	GPIO2	C7	LDATA17	E7	R/W# (WR#)	G7	D/C#	J7	RESET#
A8	GPIO1	C8	LDATA21	E8	D[4]	G8	D[2]	J8	VSS
A9	GPIO0	C9	XTAL_IN	E9	VDDPLL	G9	D[1]	J9	VDDIO
B1	VDDLCD	D1	VSS	F1	LDATA18	H1	D[21]		
B2	LDATA14	D2	LDATA8	F2	LDATA19	H2	D[17]		
B3	LDATA13	D3	LDATA7	F3	LDATA0	H3	D[15]		
B4	GAMAS0	D4	LDATA6	F4	D[19]	H4	D[12]		
B5	PWM	D5	LDATA5	F5	VSS	H5	D[9]		
B6	LLINE	D6	LFRAME	F6	VDDD	H6	D[6]		
B7	GPIO3	D7	LDATA23	F7	E(RD#)	H7	CS#		
B8	LDATA20	D8	LDATA22	F8	D[3]	H8	CONF		
B9	XTAL_OUT	D9	VSSPLL	F9	CLK	H9	D[0]		

5.2 128 pins LQFP

Figure 5-2 : Pinout Diagram – LQFP (Topview)

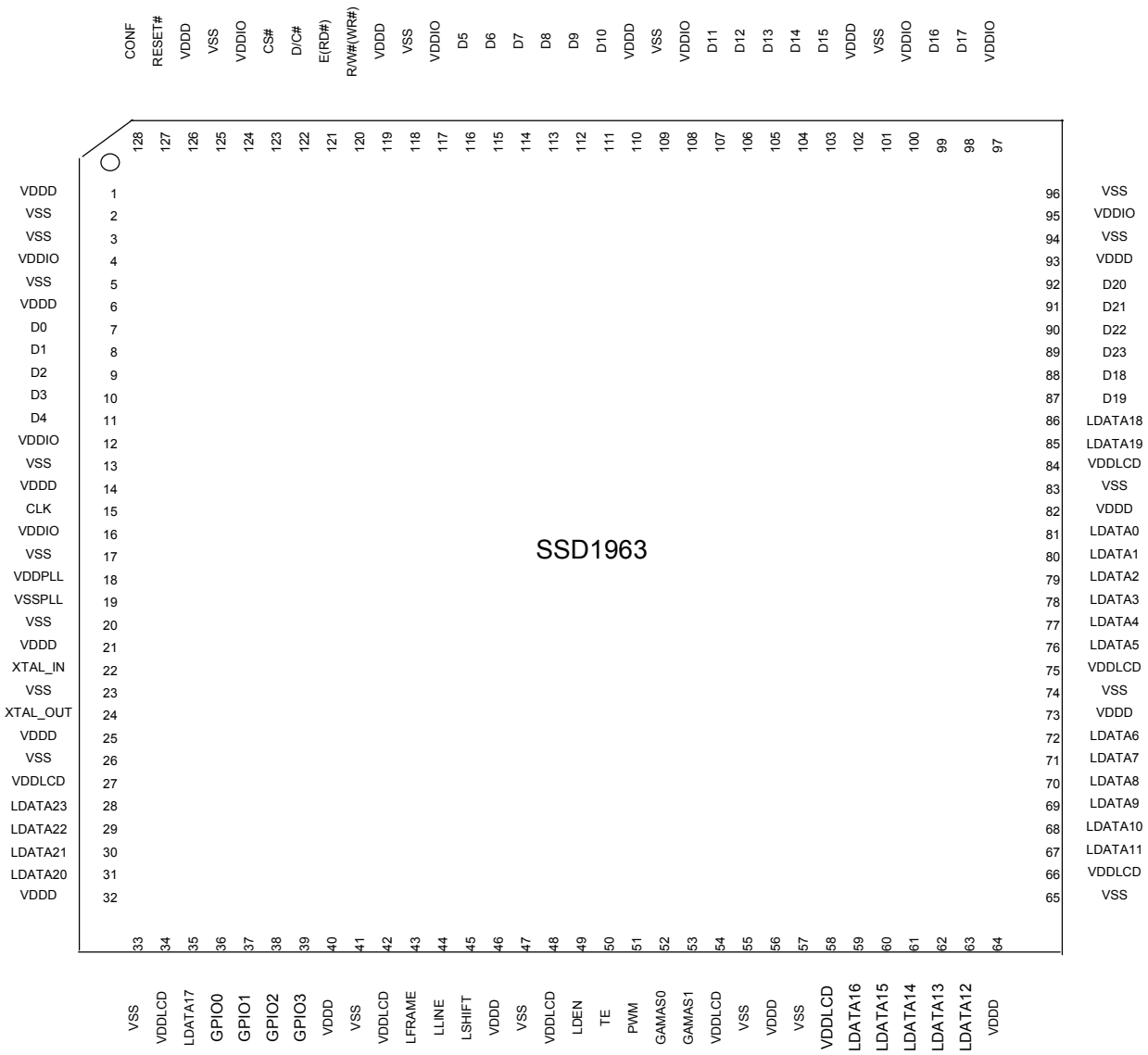


Table 5-2 : LQFP Pin Assignment Table

Pin #	Signal Name	Pin #	Signal Name	Pin #	Signal Name	Pin #	Signal Name
1	VDDD	33	VSS	65	VSS	97	VDDIO
2	VSS	34	VDDLCD	66	VDDLCD	98	D17
3	VSS	35	LDATA17	67	LDATA11	99	D16
4	VDDIO	36	GPIO0	68	LDATA10	100	VDDIO
5	VSS	37	GPIO1	69	LDATA9	101	VSS
6	VDDD	38	GPIO2	70	LDATA8	102	VDDD
7	D0	39	GPIO3	71	LDATA7	103	D15
8	D1	40	VDDD	72	LDATA6	104	D14
9	D2	41	VSS	73	VDDD	105	D13
10	D3	42	VDDLCD	74	VSS	106	D12
11	D4	43	LFRAME	75	VDDLCD	107	D11
12	VDDIO	44	LLINE	76	LDATA5	108	VDDIO
13	VSS	45	LSHIFT	77	LDATA4	109	VSS
14	VDDD	46	VDDD	78	LDATA3	110	VDDD
15	CLK	47	VSS	79	LDATA2	111	D10
16	VDDIO	48	VDDLCD	80	LDATA1	112	D9
17	VSS	49	LDEN	81	LDATA0	113	D8
18	VDDPLL	50	TE	82	VDDD	114	D7
19	VSSPLL	51	PWM	83	VSS	115	D6
20	VSS	52	GAMAS0	84	VDDLCD	116	D5
21	VDDD	53	GAMAS1	85	LDATA19	117	VDDIO
22	XTAL_IN	54	VDDLCD	86	LDATA18	118	VSS
23	VSS	55	VSS	87	D19	119	VDDD
24	XTAL_OUT	56	VDDD	88	D18	120	R/W#(WR#)
25	VDDD	57	VSS	89	D23	121	E(RD#)
26	VSS	58	VDDLCD	90	D22	122	D/C#
27	VDDLCD	59	LDATA16	91	D21	123	CS#
28	LDATA23	60	LDATA15	92	D20	124	VDDIO
29	LDATA22	61	LDATA14	93	VDDD	125	VSS
30	LDATA21	62	LDATA13	94	VSS	126	VDDD
31	LDATA20	63	LDATA12	95	VDDIO	127	RESET#
32	VDDD	64	VDDD	96	VSS	128	CONF

6 PIN DESCRIPTIONS

Key:

I = Input
O =Output
IO = Bi-directional (input/output)
P = Power pin
Hi-Z = High impedance

Table 6-1: MCU Interface Pin Mapping

Pin Name	Type	Reference Voltage Level	TFBGA Pin #	LQFP Pin #	Description
CLK	I	VDDIO	F9	15	TTL clock input. This pin should be tied to VSS if TTL clock input is not used
XTAL_IN	I	-	C9	22	Crystal oscillator input. This pin should be tied to VSS if not used
XTAL_OUT	O	-	B9	24	Crystal oscillator output. This pin should be floating if not used
CS#	I	VDDIO	H7	123	Chip select
D/C#	I	VDDIO	G7	122	Data/Command select
E(RD#)	I	VDDIO	F7	121	6800 mode: E (enable signal) 8080 mode: RD# (read strobe signal)
R/W#(WR#)	I	VDDIO	E7	120	6800 mode: R/W# 0: Write cycle 1: Read cycle 8080 mode: WR# (write strobe signal)
D[23:0]	IO	VDDIO	E8, F4, F8, G1, G2, G3, G4, G5, G6, G8, G9, H1, H2, H3, H4, H5, H6, H9, J1, J2,J3, J4, J5, J6	7, 8, 9, 10, 11, 87, 88, 89, 90, 91, 92, 98, 99, 103, 104, 105, 106, 107, 111, 112, 113, 114, 115, 116	Data bus. Pins not used should be floating
TE	O	VDDLCD	A5	50	Tearing effect

Table 6-2: LCD Interface Pin Mapping

Pin Name	Type	Reference Voltage Level	TFBGA Pin #	LQFP Pin #	Description
LFRAME	O	VDDLCD	D6	43	Vertical sync (Frame pulse)
LLINE	O	VDDLCD	B6	44	Horizontal sync (Line pulse)
LSHIFT	O	VDDLCD	A6	45	Pixel clock (Pixel shift signal)
LDEN	O	VDDLCD	C6	49	Data valid
LDATA[23:0]	O	VDDLCD	A2, A3, B2, B3, B8, C2, C3, C4, C5, C7, C8, D2, D3, D4, D5, D7, D8, E1, E2, E3, E4, F1, F2, F3	28, 29, 30, 31, 35, 59, 60, 61, 62, 63, 67, 68, 69, 70, 71, 72, 76, 77, 78, 79, 80, 81, 85, 86	RGB data
GPIO[3:0]	IO	VDDLCD	A7, A8, A9, B7	36, 37, 38, 39	These pins can be configured for display miscellaneous signals or as general purpose I/O. Default as input
GAMAS [1:0]	O	VDDLCD	A4, B4	52, 53	Gamma selection for panel
PWM	O	VDDLCD	B5	51	PWM output for backlight driver

Table 6-3: Control Signal Pin Mapping

Pin Name	Type	Reference Voltage Level	TFBGA Pin #	LQFP Pin #	Description
RESET#	I	VDDIO	J7	127	Master synchronize reset
CONF	I	VDDIO	H8	128	MCU interface configuration 0: 6800 Interface 1: 8080 Interface

Table 6-4: Power Pin Mapping

Pin Name	Type	TFBGA Pin #	LQFP Pin #	Description
VDDD	P	E6, F6	1, 6, 14, 21, 25, 32, 40, 46, 56, 64, 73, 82, 93, 102, 110, 119, 126	Power supply for internal digital circuit
VDDLCD	P	B1, C1	27, 34, 42, 48, 54, 58, 66, 75, 84	Power supply for LCD interface related pads
VDDPLL	P	E9	18	Power supply for internal analog circuit and analog I/O pads
VDDIO	P	J9	4, 12, 16, 95, 97, 100, 108, 117, 124	Power supply for digital I/O pads
VSS	P	D1, E5, F5, J8	2, 3, 5, 13, 17, 20, 23, 26, 33, 41, 47, 55, 57, 65, 74, 83, 94, 96, 101, 109, 118, 125	Ground for internal digital circuit
VSSPLL	P	D9	19	Ground for internal analog circuit and analog I/O pads

Table 6-5 : LCD Interface Pin Mapping

Pin Names	24-bit	18-bit	8-bit serial
LFRAME	FRAME		
LLINE	LINE		
LSHIFT	SHIFT		
LDEN	DEN		
LDATA23	R7	Drive 0	Drive 0
LDATA22	R6	Drive 0	Drive 0
LDATA21	R5	Drive 0	Drive 0
LDATA20	R4	Drive 0	Drive 0
LDATA19	R3	Drive 0	Drive 0
LDATA18	R2	Drive 0	Drive 0
LDATA17	R1	R5	Drive 0
LDATA16	R0	R4	Drive 0
LDATA15	G7	R3	Drive 0
LDATA14	G6	R2	Drive 0
LDATA13	G5	R1	Drive 0
LDATA12	G4	R0	Drive 0
LDATA11	G3	G5	Drive 0
LDATA10	G2	G4	Drive 0
LDATA9	G1	G3	Drive 0
LDATA8	G0	G2	Drive 0
LDATA7	B7	G1	D7
LDATA6	B6	G0	D6
LDATA5	B5	B5	D5
LDATA4	B4	B4	D4
LDATA3	B3	B3	D3
LDATA2	B2	B2	D2
LDATA1	B1	B1	D1
LDATA0	B0	B0	D0

Note

(1) These pin mappings use signal names commonly used for each panel type, however signal names may differ between panel manufacturers.

7 FUNCTIONAL BLOCK DESCRIPTIONS

7.1 MCU Interface

The MCU interface connects the MCU and SSD1963 graphics controller. The MCU interface can be configured as 6800 mode and 8080 mode by the CONF pin. By pulling the CONF pin to VSSIO, the MCU interface will be configured as 6800 mode interface. If the CONF pin is connected to VDDIO, the MCU interface will be configure in 8080 mode.

7.1.1 6800 Mode

The 6800 mode MCU interface consist of CS#, D/C#, E, R/W#, D[23:0], and TE signals (Please refer to Table 6-1 for pin multiplexed with 8080 mode). This interface supports both fixed E and clock E scheme to define a read/write cycle. If the E signal is kept high and used as enable signal, the CS# signal acts as a bus clock, the data or command will be latched into the system at the rising edge of CS#. If the user wants to use the E pin as the clock pin, the CS# pin then need to be fixed to logic 0 to select the chip. Then the falling edge of the E signal will latch the data or command. For details, please refer to the timing diagram in chapter 13.2.1.

7.1.2 8080 Mode

The 8080 mode MCU interface consist of CS#, D/C#, RD#, WR#, D[23:0] and TE signals (Please refer to Table 6-1 for pin multiplexed with 6800 mode). This interface use WR# to define a write cycle and RD# for read cycle. If the WR# goes low when the CS# signal is low, the data or command will be latched into the system at the rising edge of WR#. Similarly, the read cycle will start when RD# goes low and end at the rising edge of RD#. The detailed timing will show in the chapter 13.2.2.

7.1.3 Register Pin Mapping

When user access the registers via the parallel MCU interface, only D[7:0] will be used regardless the width of the pixel data is. Therefore, D[23:8] will only be used to address the display data only. This provided the possibility that the pixel data format as shown in Table 7-1 can be configured by command 0xF0.

7.1.4 Pixel Data Format

Both 6800 and 8080 support 8-bit, 9-bit, 16-bit, 18-bit and 24-bit data bus. Depending on the width of the data bus, the display data are packed into the data bus in different ways.

Table 7-1: Pixel Data Format

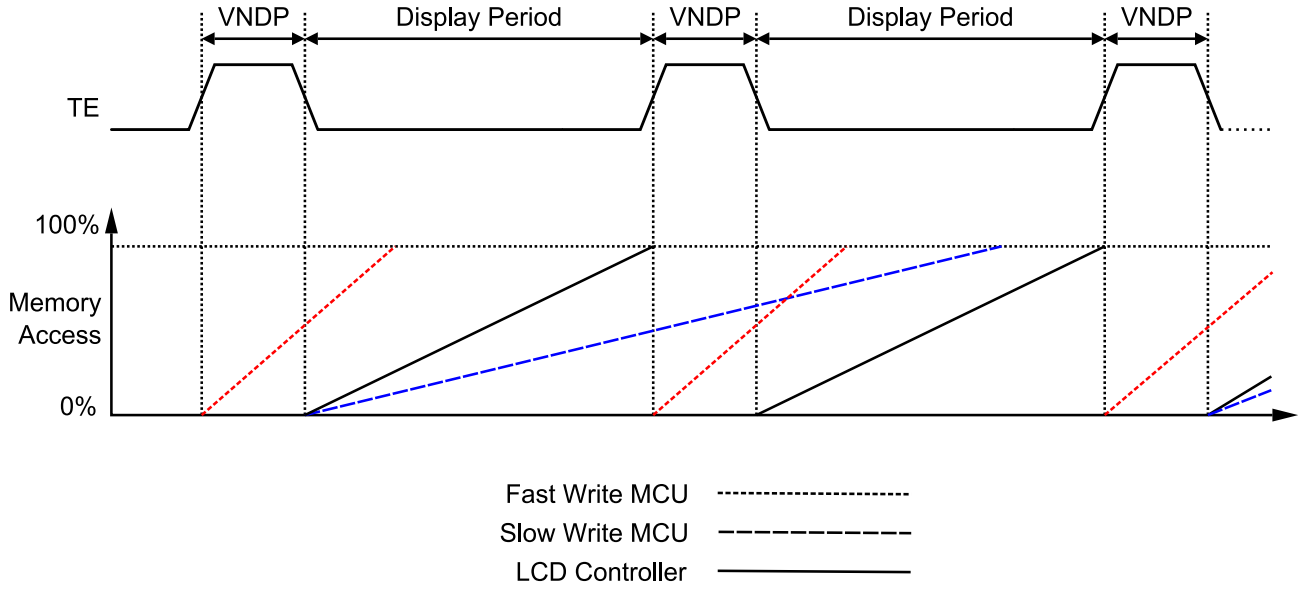
Interface	Cycle	D[23]	D[22]	D[21]	D[20]	D[19]	D[18]	D[17]	D[16]	D[15]	D[14]	D[13]	D[12]	D[11]	D[10]	D[9]	D[8]	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
24 bits	1st	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
18 bits	1st							R5	R4	R3	R2	R1	R0	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1	B0
16 bits (565 format)	1st									R5	R4	R3	R2	R1	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1
16 bits	1st									R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0
	2nd									B7	B6	B5	B4	B3	B2	B1	B0	R7	R6	R5	R4	R3	R2	R1	R0
	3rd									G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
12 bits	1st													R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4
	2nd													G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0
9 bits	1st																R5	R4	R3	R2	R1	R0	G5	G4	G3
	2nd																G2	G1	G0	B5	B4	B3	B2	B1	B0
8 bits	1st																	R7	R6	R5	R4	R3	R2	R1	R0
	2nd																	G7	G6	G5	G4	G3	G2	G1	G0
	3rd																	B7	B6	B5	B4	B3	B2	B1	B0

7.1.5 Tearing Effect Signal (TE)

The Tearing Effect Signal (TE) is a feedback signal from the LCD Controller to MCU. This signal reveals the display status of LCD controller. In the non-display period, the TE signal will go high. Therefore, this signal enables the MCU to send data by observing the non-display period to avoid tearing.

Figure 7-1 shows how the TE signal helps to avoid tearing. If the MCU writing speed is slower than the display speed, the display data should be updated after the LCD controller start to scan the frame buffer. Then the LCD controller will always display the old memory content until the next frame. However, if the MCU is faster than the LCD controller, it should start updating the display content in the vertical non-display period (VNDP) to enable the LCD controller will always get the newly updated data.

Figure 7-1: Relationship between Tearing Effect Signal and MCU Memory Writing



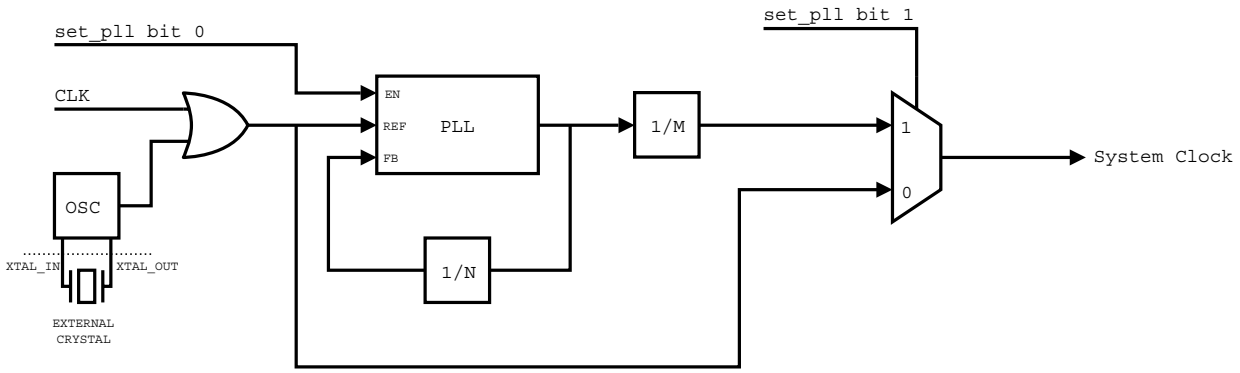
In SSD1963, users can configure the TE signal to reflect the vertical non-display period only or reflect both vertical and horizontal non-display period. With the additional horizontal non-display period information, the MCU can control the refresh action in more accurately by counting the horizontal line scanned by the LCD controller. Usually, a fast MCU will not need horizontal non-display period. But a slow MCU will need it to ensure the frame buffer update process always lags behind the LCD controller.

7.2 System Clock Generation

The system clock of SSD1963 is generated by the built-in PLL. The reference clock of the PLL can come from either the CLK pin or the external crystal oscillator. Since the CLK pin and the output of the oscillator was connected to PLL with an "OR" gate, the unused clock must be tied to VSS.

Before the PLL output is configured as the system clock by the bit 1 of "set_pll" command 0xE0, the system will be clocked by the reference clock. This enables the user to send the "set_pll_mn" command 0xE2 to the PLL for frequency configuration. When the PLL frequency is configured and the PLL was enabled with the bit 0 of "set_pll" command 0xE0, the user should still wait for 100ms for the PLL to lock. Then the PLL is ready and can be configured as system clock with the bit 1 of "set_pll" command 0xE0.

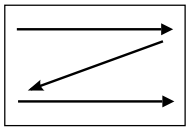
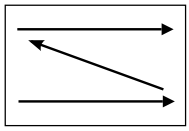
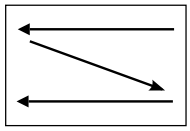
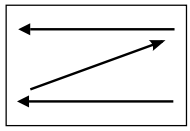
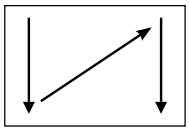
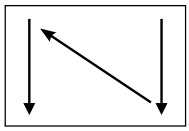
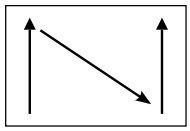
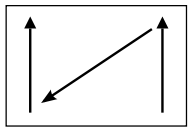
Figure 7-2: Clock Control Diagram



7.3 Frame Buffer

There are 1215K bytes built-in SRAM inside SSD1963 to use as frame buffer. When the frame buffer is written or read, the "address counter" will automatically increase by one or decrease by one depends on the frame buffer settings.

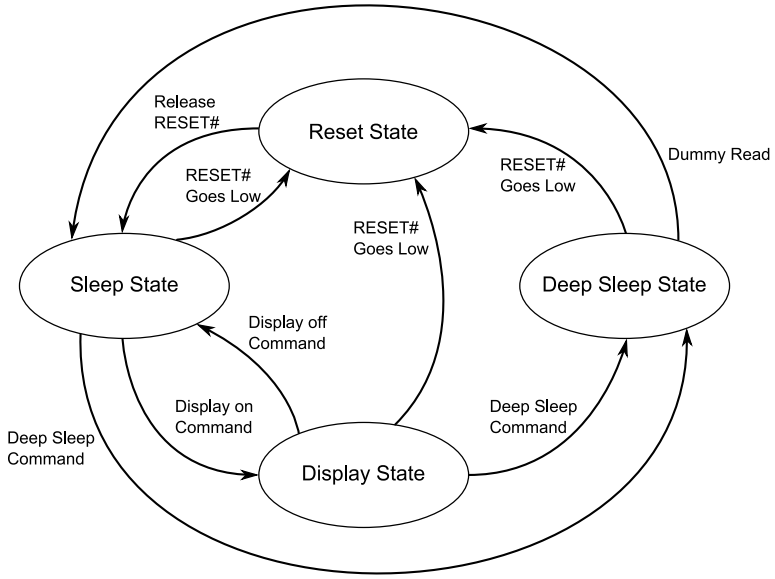
Table 7-2: Frame Buffer Settings regarding to set_address_mode command 0x36

	Option 1 Horizontal: Increment Vertical: Increment B6 = 0; B7 = 0	Option 2 Horizontal: Increment Vertical: Decrement B6 = 0; B7 = 1	Option 3 Horizontal: Decrement Vertical: Increment B6 = 1; B7 = 0	Option 4 Horizontal: Decrement Vertical: Decrement B6 = 1; B7 = 1
Horizontal Frame Buffer Mode B5 = 0	00000h  5DC00h	00000h  5DC00h	00000h  5DC00h	00000h  5DC00h
Vertical Frame Buffer Mode B5 = 1	00000h  5DC00h	00000h  5DC00h	00000h  5DC00h	00000h  5DC00h

7.4 System Clock and Reset Manager

The "System Clock and Reset Manager" distributes the reset signal and clock signal to the entire system. It controls the Clock Generator and contains clock gating circuitry to turn on and off the clock of each functional module. Also, it divides the root clock from Clock Generator to operation clocks for different module. The System Clock and Reset Manager also manage the reset signals to ensure all the module are reset to appropriate status when the system are in reset state, deep sleep state, sleep state and display state. Figure 7-3 shows a state diagram of four operation states of SSD1963.

Figure 7-3: State Diagram of SSD1963



Reset State:	Deep Sleep State:	Sleep State:	Display State:
Clock Generator Stop	Clock Generator Stop	Clock Generator On	Clock Generator On
Unable to Receive Command	Unable to Receive Command	Able to Receive Command	Able to Receive Command
Unable to Update Frame Buffer	Unable to Update Frame Buffer	Able to Update Frame Buffer	Able to Update Frame Buffer
Display Off	Display Off	Display Off	Display On
All Settings Reset	All Settings Retain	All Settings Retain	All Settings Retain

7.5 LCD Controller

7.5.1 Display Format

The LCD controller reads the frame buffer and generates display signals according to the selected display panel format. SSD1963 supports common RAM-less TFT driver using generic RGB data format.

7.5.2 General Purpose Input/Output (GPIO)

The GPIO pins can operate in 2 modes, GPIO mode and miscellaneous display signal mode. When the pins are configured as GPIOs, these pins can be controlled directly by MCU. Therefore, user can use these pins to emulate other interface such as SPI or I2C. If these pins are configured as display signals, they will toggle with display periodically according to the signal settings. They can be set to toggle once a frame, once a line or in arbitrary period. Therefore they can be configured as some common signal needed for different panels such as STH or LP.

8 COMMAND TABLE

Hex Code	Command	Description
0x00	nop	No operation
0x01	soft_reset	Software Reset
0x0A	get_power_mode	Get the current power mode
0x0B	get_address_mode	Get the frame buffer to the display panel read order
0x0C	Reserved	Reserved
0x0D	get_display_mode	The SSD1963 returns the Display Image Mode.
0x0E	get_tear_effect_status	Get the Tear Effect status
0x0F	Reserved	Reserved
0x10	enter_sleep_mode	Turn off the panel. This command will pull low the GPIO0. If GPIO0 is configured as normal GPIO or LCD miscellaneous signal with command set_gpio_conf, this command will be ignored.
0x11	exit_sleep_mode	Turn on the panel. This command will pull high the GPIO0. If GPIO0 is configured as normal GPIO or LCD miscellaneous signal with command set_gpio_conf, this command will be ignored.
0x12	enter_partial_mode	Part of the display area is used for image display.
0x13	enter_normal_mode	The whole display area is used for image display.
0x20	exit_invert_mode	Displayed image colors are not inverted.
0x21	enter_invert_mode	Displayed image colors are inverted.
0x26	set_gamma_curve	Selects the gamma curve used by the display panel.
0x28	set_display_off	Blanks the display panel
0x29	set_display_on	Show the image on the display panel
0x2A	set_column_address	Set the column address
0x2B	set_page_address	Set the page address
0x2C	write_memory_start	Transfer image information from the host processor interface to the SSD1963 starting at the location provided by set_column_address and set_page_address
0x2E	read_memory_start	Transfer image data from the SSD1963 to the host processor interface starting at the location provided by set_column_address and set_page_address
0x30	set_partial_area	Defines the partial display area on the display panel
0x33	set_scroll_area	Defines the vertical scrolling and fixed area on display area
0x34	set_tear_off	Synchronization information is not sent from the SSD1963 to the host processor
0x35	set_tear_on	Synchronization information is sent from the SSD1963 to the host processor at the start of VFP
0x36	set_address_mode	Set the read order from frame buffer to the display panel
0x37	set_scroll_start	Defines the vertical scrolling starting point
0x38	exit_idle_mode	Full color depth is used for the display panel
0x39	enter_idle_mode	Reduce color depth is used on the display panel.
0x3A	Reserved	Reserved
0x3C	write_memory_continue	Transfer image information from the host processor interface to the SSD1963 from the last written location
0x3E	read_memory_continue	Read image data from the SSD1963 continuing after the last read_memory_continue or read_memory_start

Hex Code	Command	Description
0x44	set_tear_scanline	Synchronization information is sent from the SSD1963 to the host processor when the display panel refresh reaches the provided scanline
0x45	get_scanline	Get the current scan line
0xA1	read_ddb	Read the DDB from the provided location
0xA8	Reserved	Reserved
0xB0	set_lcd_mode_	Set the LCD panel mode and resolution
0xB1	get_lcd_mode	Get the current LCD panel mode, pad strength and resolution
0xB4	set_hori_period	Set front porch
0xB5	get_hori_period	Get current front porch settings
0xB6	set_vert_period	Set the vertical blanking interval between last scan line and next LFRAME pulse
0xB7	get_vert_period	Set the vertical blanking interval between last scan line and next LFRAME pulse
0xB8	set_gpio_conf	Set the GPIO configuration. If the GPIO is not used for LCD, set the direction. Otherwise, they are toggled with LCD signals.
0xB9	get_gpio_conf	Get the current GPIO configuration
0xBA	set_gpio_value	Set GPIO value for GPIO configured as output
0xBB	get_gpio_status	Read current GPIO status. If the individual GPIO was configured as input, the value is the status of the corresponding pin. Otherwise, it is the programmed value.
0xBC	set_post_proc	Set the image post processor
0xBD	get_post_proc	Set the image post processor
0xBE	set_pwm_conf	Set the image post processor
0xBF	get_pwm_conf	Set the image post processor
0xC0	set_lcd_gen0	Set the rise, fall, period and toggling properties of LCD signal generator 0
0xC1	get_lcd_gen0	Get the current settings of LCD signal generator 0
0xC2	set_lcd_gen1	Set the rise, fall, period and toggling properties of LCD signal generator 1
0xC3	get_lcd_gen1	Get the current settings of LCD signal generator 1
0xC4	set_lcd_gen2	Set the rise, fall, period and toggling properties of LCD signal generator 2
0xC5	get_lcd_gen2	Get the current settings of LCD signal generator 2
0xC6	set_lcd_gen3	Set the rise, fall, period and toggling properties of LCD signal generator 3
0xC7	get_lcd_gen3	Get the current settings of LCD signal generator 3
0xC8	set_gpio0_rop	Set the GPIO0 with respect to the LCD signal generators using ROP operation. No effect if the GPIO0 is configured as general GPIO.
0xC9	get_gpio0_rop	Get the GPIO0 properties with respect to the LCD signal generators.
0xCA	set_gpio1_rop	Set the GPIO1 with respect to the LCD signal generators using ROP operation. No effect if the GPIO1 is configured as general GPIO.
0xCB	get_gpio1_rop	Get the GPIO1 properties with respect to the LCD signal generators.
0xCC	set_gpio2_rop	Set the GPIO2 with respect to the LCD signal generators using ROP operation. No effect if the GPIO2 is configured as general GPIO.

Hex Code	Command	Description
0xCD	get_gpio2_rop	Get the GPIO2 properties with respect to the LCD signal generators.
0xCE	set_gpio3_rop	Set the GPIO3 with respect to the LCD signal generators using ROP operation. No effect if the GPIO3 is configured as general GPIO.
0xCF	get_gpio3_rop	Get the GPIO3 properties with respect to the LCD signal generators.
0xD0	set_dbc_conf	Set the dynamic back light configuration
0xD1	get_dbc_conf	Get the current dynamic back light configuration
0xD4	set_dbc_th	Set the threshold for each level of power saving
0xD5	get_dbc_th	Get the threshold for each level of power saving
0xE0	set_pll	Start the PLL. Before the start, the system was operated with the crystal oscillator or clock input
0xE2	set_pll_mn	Set the PLL
0xE3	get_pll_mn	Get the PLL settings
0xE4	get_pll_status	Get the current PLL status
0xE5	set_deep_sleep	Set deep sleep mode
0xE6	set_lshift_freq	Set the LSHIFT (pixel clock) frequency
0xE7	get_lshift_freq	Get current LSHIFT (pixel clock) frequency setting
0xE8	Reserved	Reserved
0xE9	Reserved	Reserved
0xF0	set_pixel_data_interface	Set the pixel data format of the parallel host processor interface
0xF1	get_pixel_data_interface	Get the current pixel data format settings
0xFF	Reserved	Reserved

9 COMMAND DESCRIPTIONS

9.1 nop

Command 0x00
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	0	0	0	0	0	00

Description
No operation.

9.2 soft_reset

Command 0x01
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	0	0	0	0	1	01

Description
The SSD1963 performs a software reset. All the configuration register will be reset except command 0xE0 to 0xE5.

Note :

The host processor must wait 5ms before sending any new commands to a SSD1963 following this command.

9.3 get_power_mode

Command 0x0A
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	0	1	0	1	0	0A
Parameter 1	1	0	A_6	A_5	A_4	A_3	A_2	0	0	xx

Description
Get the current power mode

A[6] : Idle mode on/off (POR = 0)
0 Idle mode off
1 Idle mode on

A[5] : Partial mode on/off (POR = 0)
0 Partial mode off
1 Partial mode on

A[4] : Sleep mode on/off (POR = 0)
0 Sleep mode on
1 Sleep mode off

A[3] : Display normal mode on/off (POR = 1)
0 Display normal mode off

1 Display normal mode on (partial mode and vertical scroll off)

A[2] : Display on/off (POR = 0)
0 Display is off
1 Display is on

9.4 get_address_mode

Command 0x0B
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	0	1	0	1	1	0B
Parameter 1	1	A_7	A_6	A_5	A_4	A_3	A_2	0	0	xx

Description
Get the frame buffer to the display panel read order

A[7] : Page address order (POR = 0)
0 Top to bottom
1 Bottom to top

A[6] : Column address order (POR = 0)
0 Left to right
1 Right to left

A[5] : Page / Column order (POR = 0)
0 Normal mode
1 Reverse mode

A[4] : Line address order (POR = 0)
0 LCD refresh top to bottom
1 LCD refresh bottom to top

A[3] : RGB / BGR order (POR = 0)
0 RGB
1 BGR

A[2] : Display data latch data (POR = 0)
0 LCD refresh left to right
1 LCD refresh right to left

9.5 get_display_mode

Command 0x0D
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	0	1	1	0	1	0D
Parameter 1	1	A_7	0	A_5	0	0	A_2	A_1	A_0	xx

Description
Get the Display Image Mode status.

A[7] : Vertical scrolling on/off (POR = 0)
0 Vertical scrolling is off

1 Vertical scrolling is on

A[5] : Invert mode on/off (POR = 0)
0 Inversion is off
1 Inversion is on

A[2:0] : Gamma curve selection (POR = 011)
000 Gamma curve 0
001 Gamma curve 1
010 Gamma curve 2
011 Gamma curve 3
100 Reserved
101 Reserved
110 Reserved
111 Reserved

9.6 get_tear_effect_status

Command 0x0E
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	0	1	1	1	0	0E
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx

Description
Get the current Tear Effect mode from the SSD1963

A[7] : Tearing effect line mode (POR = 0)
0 Tearing effect off
1 Tearing effect on

9.7 enter_sleep_mode

Command 0x10
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	1	0	0	0	0	10

Description
Turn off the panel. This command causes the SSD1963 to enter sleep mode and pull low the GPIO[0] if set_gpio_conf (0xB8) B0 = 0
If GPIO[0] is configured as normal GPIO or LCD miscellaneous signal with command set_gpio_conf (0xB8), this command will not affect the GPIO[0].

Note :

The host processor must wait 5ms before sending any new commands to a SSD1963 following this command.

9.8 exit_sleep_mode

Command 0x11
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	1	0	0	0	1	11

Description
Turn on the panel. This command causes the SSD1963 to exit sleep mode and will pull high the GPIO[0] if set_gpio_conf (0xB8) B0 = 0.
If GPIO[0] is configured as normal GPIO or LCD miscellaneous signal with command set_gpio_conf (0xB8), this command will not affect the GPIO[0].

Note :
The host processor must wait 5ms after sending this command before sending another command.
**This command will automatic trigger set_display_on (0x29)

9.9 enter_partial_mode

Command 0x12
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	1	0	0	1	0	12

Description
Once enter_partial_mode is triggered, the Partial Display Mode window is described by the set_partial_area (0x30).
Once enter_normal_mode (0x13) is triggered, partial display mode will end.

9.10 enter_normal_mode

Command 0x13
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	1	0	0	1	1	13

Description
This command causes the SSD1963 to enter the normal mode. Normal mode is defined as partial display and vertical scroll mode are off. That means the whole display area is used for image display.

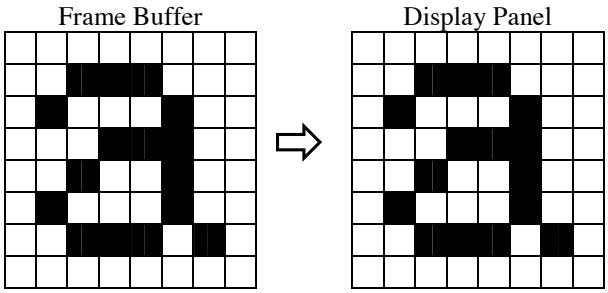
9.11 exit_invert_mode

Command 0x20
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	0	0	0	0	20

Description
This command causes the SSD1963 to stop inverting the image data on the display panel. The frame buffer contents remain unchanged.

Figure 9-1: Exit Invert mode example



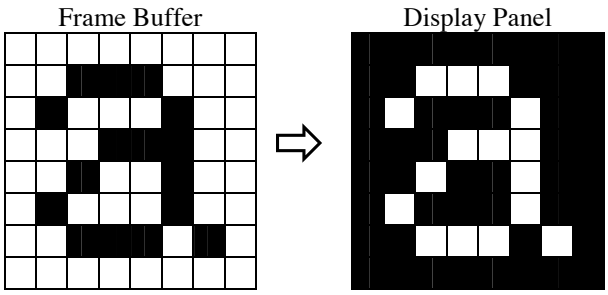
9.12 enter_invert_mode

Command 0x21
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	0	0	0	1	21

Description
This command causes the SSD1963 to invert the image data only on the display panel. The frame buffer contents remain unchanged.

Figure 9-2: Enter Invert mode example



9.13 set_gamma_curve

Command 0x26
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	0	1	1	0	26
Parameter 1	1	0	0	0	0	A_3	A_2	A_1	A_0	xx

Description
Selects the gamma curve used by the display panel.

A[3:0]	Gamma curve selection (POR = 1000)	GAMAS[1]	GAMAS[0]
0000	No gamma curve selected (Same as 0001b)	0	0
0001	Gamma curve 0	0	0
0010	Gamma curve 1	0	1
0100	Gamma curve 2	1	0
1000	Gamma curve 3	1	1
Others	Reserved		

9.14 set_display_off

Command 0x28
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	1	0	0	0	28

Description
Blanks the display panel. The frame buffer contents remain unchanged.

9.15 set_display_on

Command 0x29
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	1	0	0	1	29

Description
Show the image on the display panel

9.16 set_column_address

Command 0x2A
Parameters 4

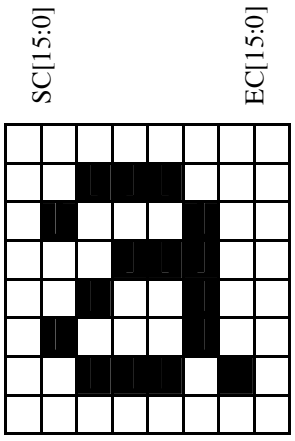
	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	1	0	1	0	2A
Parameter 1	1	SC_{15}	SC_{14}	SC_{13}	SC_{12}	SC_{11}	SC_{10}	SC_9	SC_8	xx
Parameter 2	1	SC_7	SC_6	SC_5	SC_4	SC_3	SC_2	SC_1	SC_0	xx
Parameter 3	1	EC_{15}	EC_{14}	EC_{13}	EC_{12}	EC_{11}	EC_{10}	EC_9	EC_8	xx
Parameter 4	1	EC_7	EC_6	EC_5	EC_4	EC_3	EC_2	EC_1	EC_0	xx

Description
Set the column address of frame buffer accessed by the host processor with the read_memory_continue (0x3E) and write_memorty_continue (0x3C)..

SC[15:8] : Start column number high byte (POR = 00000000)
SC[7:0] : Start column number low byte (POR = 00000000)
EC[15:8] : End column number high byte (POR = 00000000)
EC[7:0] : End column number low byte (POR = 00000000)

Note : SC[15:0] must always be equal to or less than EC[15:0]

Figure 9-3: Set Column Address example



9.17 set_page_address

Command 0x2B
Parameters 4

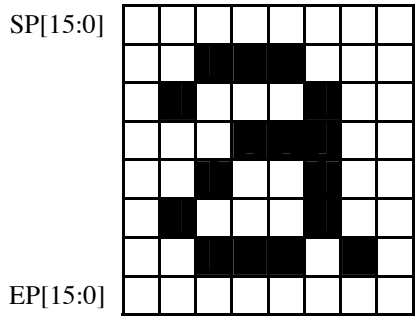
	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	1	0	0	1	2B
Parameter 1	1	SP_{15}	SP_{14}	SP_{13}	SP_{12}	SP_{11}	SP_{10}	SP_9	SP_8	xx
Parameter 2	1	SP_7	SP_6	SP_5	SP_4	SP_3	SP_2	SP_1	SP_0	xx
Parameter 3	1	EP_{15}	EP_{14}	EP_{13}	EP_{12}	EP_{11}	EP_{10}	EP_9	EP_8	xx
Parameter 4	1	EP_7	EP_6	EP_5	EP_4	EP_3	EP_2	EP_1	EP_0	xx

Description
Set the page address of the frame buffer accessed by the host processor with the read_memory_start (0x2C), write_memory_start (0x2E), read_memory_continue (0x3E) and write_memory_continue (0x3C)..

SP[15:8] : Start page (row) number high byte (POR = 00000000)
SP[7:0] : Start page (row) number low byte (POR = 00000000)
EP[15:8] : End page (row) number high byte (POR = 00000000)
EP[7:0] : End page (row) number low byte (POR = 00000000)

Note : SP[15:0] must always be equal to or less than EP[15:0]

Figure 9-4: Set Page Address example



9.18 write_memory_start

Command 0x2C
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	1	1	0	0	2C

Description
Transfer image information from the host processor interface to the SSD1963 starting at the location provided by set_column _address (0x2A) and set _page_address (0x2B).

If set_address_mdoe (0x36) A[5] = 0:

The column and page address are reset to the Start Column (SC) and Start Page (SP), respectively.

Pixel Data 1 is stored in frame buffer at (SC, SP). The column address is then incremented and pixels are written to the frame buffer until the column address equals the End Column (EC) value. The column address is then reset to SC and the page address is incremented. Pixels are written to the frame buffer until the page address equals the End Page (EP) value and the column address equals the EC value, or the host processor sends another command. If the number of pixels exceeds (EC – SC + 1) * (EP – SP + 1) the extra pixels are ignored.

If set_address_mode (0x36) A[5] = 1:

The column and page address are reset to the Start Column (SC) and Start Page (SP), respectively.

Pixel Data 1 is stored in frame buffer at (SC, SP). The page address is then incremented and pixels are written to the frame buffer until the page address equals the End Page (EP) value. The page address is then reset to SP and the column address is incremented. Pixels are written to the frame buffer until the column address equals the End column (EC) value and the page address equals the EP value, or the host processor sends another command. If the number of pixels exceeds (EC – SC + 1) * (EP – SP + 1) the extra pixels are ignored.

9.19 read_memory_start

Command 0x2E
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	0	1	1	1	0	2E

Description
Transfer image data from the SSD1963 to the host processor interface starting at the location provided by set_column_address (0x2A) and set_page_address (0x2B).

If set_address_mode A[5] = 0:

The column and page address are reset to the Start Column (SC) and Start Page (SP), respectively.

Pixels Data 1 are read from frame buffer at (SC, SP). The column address is then incremented and pixels read from the frame buffer until the column address equals the End Column (EC) value. The column address is then reset to SC and the page address is incremented. Pixels are read from the frame buffer until the page address equals the End Page (EP) value and the column address equals the EC value, or the host processor sends another command.

If set_address_mode (0x36) A[5] = 1:

The column and page address are reset to the Start Column (SC) and Start Page (SP), respectively.

Pixels Data 1 are read from frame buffer at (SC, SP). The page address is then incremented and pixels read from the frame buffer until the page address equals the End Page (EP) value. The page address is then reset to SP and the column address is incremented. Pixels are read from the frame buffer until the column address equals the End Column (EC) value and the page address equals the EP value, or the host processor sends another command.

9.20 set_partial_area

Command 0x30
Parameters 4

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	0	0	0	0	30
Parameter 1	1	SR_{15}	SR_{14}	SR_{13}	SR_{12}	SR_{11}	SR_{10}	SR_9	SR_8	xx
Parameter 2	1	SR_7	SR_6	SR_5	SR_4	SR_3	SR_2	SR_1	SR_0	xx
Parameter 3	1	ER_{15}	ER_{14}	ER_{13}	ER_{12}	ER_{11}	ER_{10}	ER_9	ER_8	xx
Parameter 4	1	ER_7	ER_6	ER_5	ER_4	ER_3	ER_2	ER_1	ER_0	xx

Description
This command defines the Partial Display mode's display area. There are two parameters associated with this command, the first defines the Start Row (SR) and the second the End Row (ER). SR and ER refer to the Frame Buffer Line Pointer.

SR[15:8] : Start display row number high byte (POR = 00000000)
SR[7:0] : Start display row number low byte (POR = 00000000)
ER[15:8] : End display row number high byte (POR = 00000000)
ER[7:0] : End display row number low byte (POR = 00000000)
Note : SR[15:0] and ER[15:0] cannot be 0000h nor exceed the last vertical line number.

If End Row > Start Row

Figure 9-5: Set Partial Area with set_address_mode (0x36) A[4] = 0 when End Row > Start Row

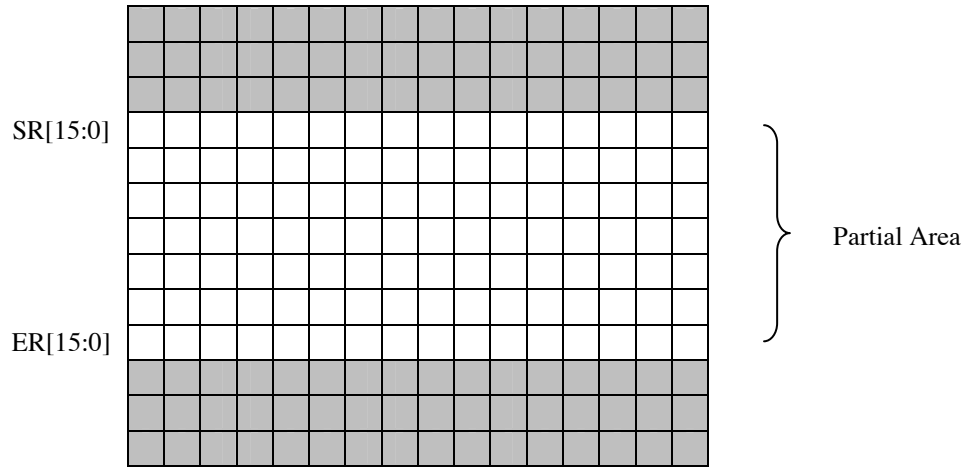
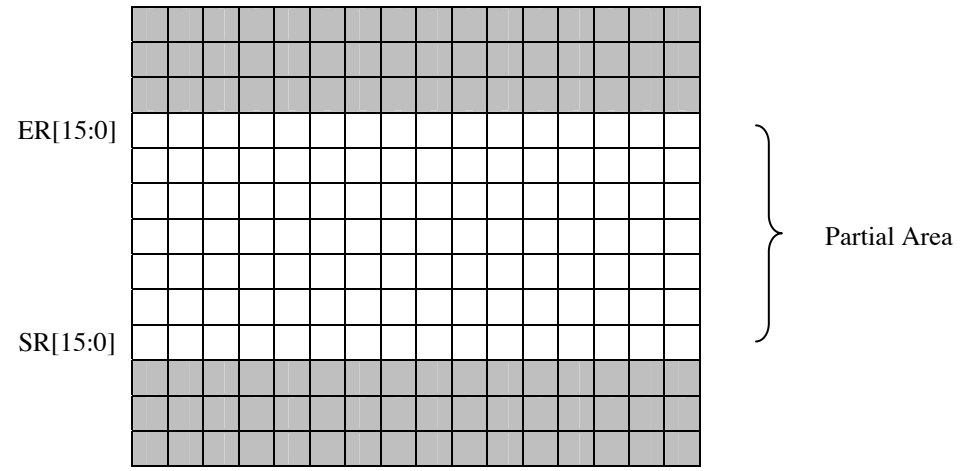


Figure 9-6: Set Partial Area with set_address_mode (0x36) A[4] = 1 when End Row > Start Row



If Start Row > End Row

Figure 9-7: Set Partial Area with set_address_mode (0x36) A[4] = 0 when Start Row > End Row

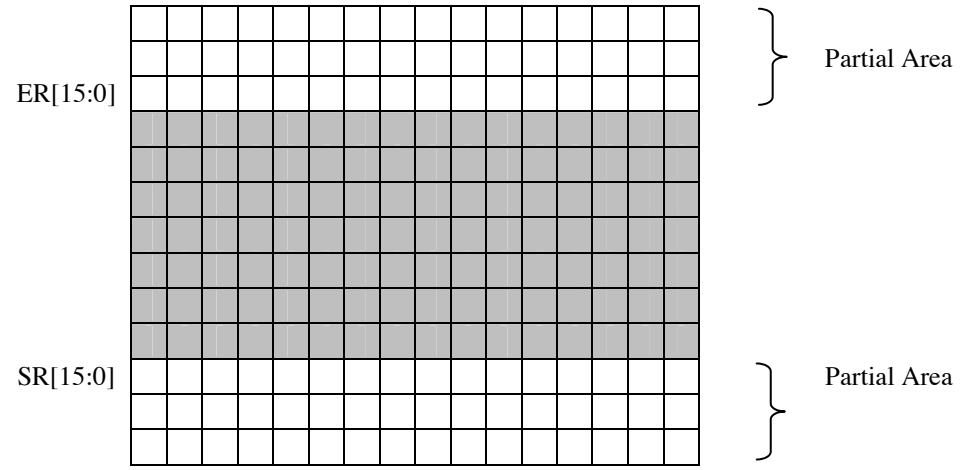
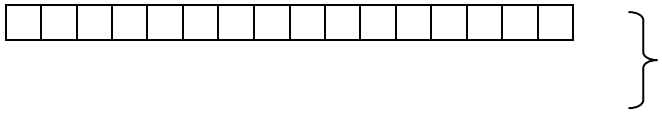
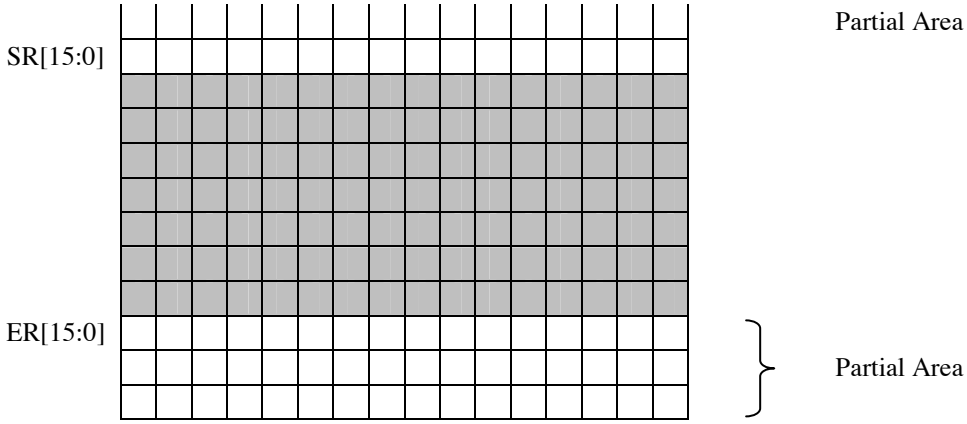


Figure 9-8: Set Partial Area with set_address_mode (0x36) A[4] = 1 when Start Row > End Row





9.21 set_scroll_area

Command 0x33
Parameters 6

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	0	0	1	1	33
Parameter 1	1	TFA_{15}	TFA_{14}	TFA_{13}	TFA_{12}	TFA_{11}	TFA_{10}	TFA_9	TFA_8	xx
Parameter 2	1	TFA_7	TFA_6	TFA_5	TFA_4	TFA_3	TFA_2	TFA_1	TFA_0	xx
Parameter 3	1	VSA_{15}	VSA_{14}	VSA_{13}	VSA_{12}	VSA_{11}	VSA_{10}	VSA_9	VSA_8	xx
Parameter 4	1	VSA_7	VSA_6	VSA_5	VSA_4	VSA_3	VSA_2	VSA_1	VSA_0	xx
Parameter 5	1	BFA_{15}	BFA_{14}	BFA_{13}	BFA_{12}	BFA_{11}	BFA_{10}	BFA_9	BFA_8	xx
Parameter 6	1	BFA_7	BFA_6	BFA_5	BFA_4	BFA_3	BFA_2	BFA_1	BFA_0	xx

Description
Defines the vertical scrolling and fixed area on display area

TFA[15:8] : High byte of Top Fixed Area number in lines from the top of the frame buffer (POR = 00000000)
TFA[7:0] : Low byte of Top Fixed Area number in lines from the top of the frame buffer (POR = 00000000)

VSA[15:8] : High byte of Vertical scrolling area in number of lines of the frame buffer (POR = 00000000)
VSA[7:0] : Low byte of Vertical scrolling area in number of lines of the frame buffer (POR = 00000000)

BFA[15:8] : High byte of Bottom Fixed Area in number of lines from the bottom of the frame buffer (POR = 00000000)
BFA[7:0] : Low byte of Bottom Fixed Area in number of lines from the bottom of the frame buffer (POR = 00000000)

If set_address_mode (0x36) A[4] = 0 :

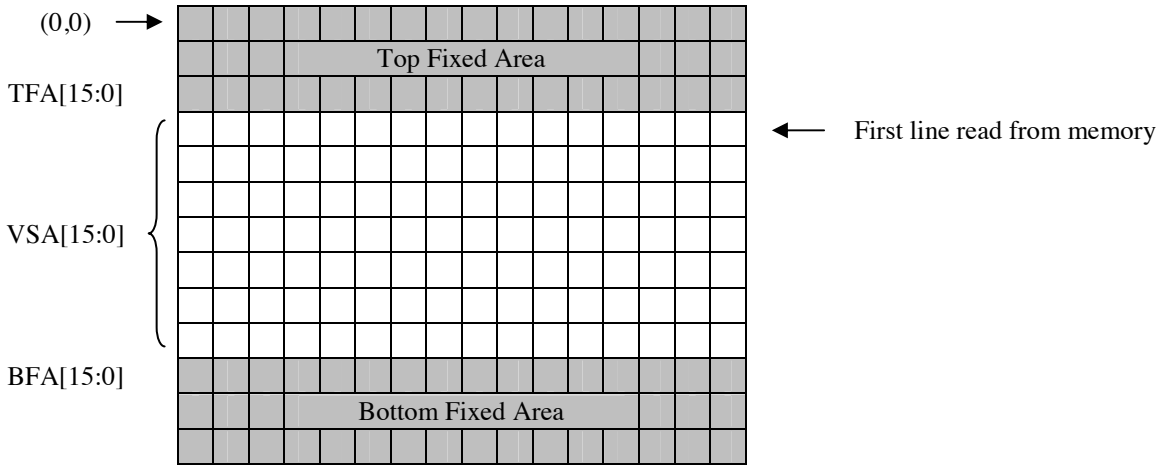
The TFA[15:0] describes the Top Fixed Area in number of lines from the top of the frame buffer. The top of the frame buffer and top of the display panel are aligned.

The VSA[15:0] describes the height of the Vertical Scrolling Area in number of lines of frame buffer from the Vertical Scrolling Start Address. The first line of the Vertical Scrolling Area starts immediately after the bottom most line of the Top Fixed Area. The last line of the Vertical Scrolling Area ends immediately before the top most line of the Bottom Fixed Area.

The BFA[15:0] describes the Bottom Fixed Area in number of lines from the bottom of the frame buffer. The bottom of the frame buffer and bottom of the display panel are aligned.

TFA, VSA and BFA refer to the Frame Buffer Line Pointer.

Figure 9-9: Set Scroll Area with set_address_mode (0x36) A[4] = 0



If set_address_mode (0x36) A[4] = 1 :

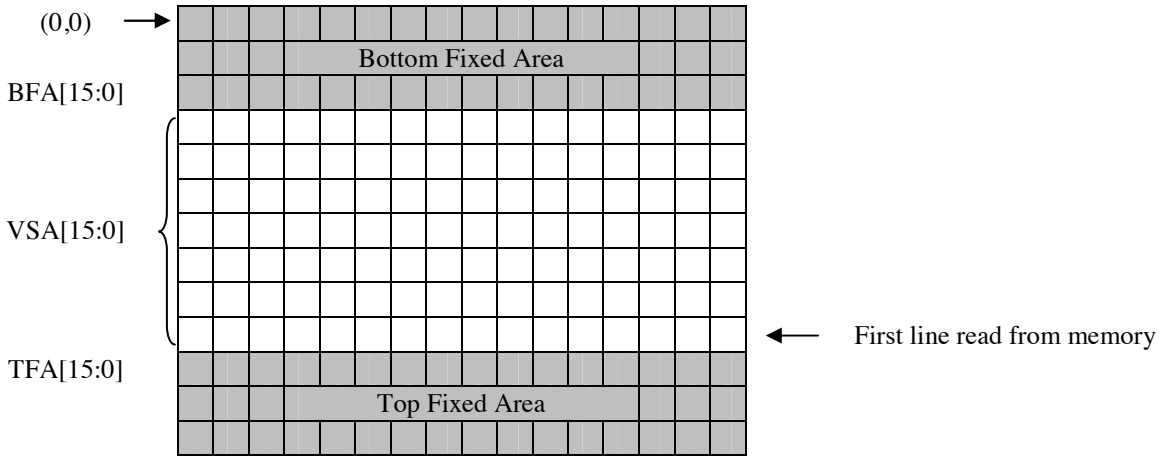
The TFA[15:0], describes the Top Fixed Area in number of lines from the bottom of the frame buffer. The bottom of the frame buffer and bottom of the display panel are aligned.

The VSA[15:0] describes the height of the Vertical Scrolling Area in number of lines of frame buffer from the Vertical Scrolling Start Address. The first line of the Vertical Scrolling Area starts immediately after the top most line of the Top Fixed Area. The last line of the Vertical Scrolling Area ends immediately before the bottom most line of the Bottom Fixed Area.

The BFA[15:0] describes the Bottom Fixed Area in number of lines from the top of the frame buffer. The top of the frame buffer and top of the display panel are aligned.

TFA, VSA and BFA refer to the Frame Buffer Line Pointer.

Figure 9-10: Set Scroll Area with set_address_mode (0x36) A[4] = 1



Note :

The sum of TFA, VSA and BFA must equal the number of the display panel's horizontal lines (pages), otherwise Scrolling mode is undefined.

In Vertical Scroll Mode, set_address_mode (0x36) A[5] should be set to '0' – this only affects the Frame Buffer Write.

9.22 set_tear_off

Command 0x34
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	0	1	0	0	34

Description
TE signal is not sent from the SSD1963 to the host processor.

9.23 set_tear_on

Command 0x35
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	0	1	0	1	35
Parameter 1	1	0	0	0	0	0	0	0	A_0	xx

Description
TE signal is sent from the SSD1963 to the host processor at the start of VFP.

A[0] : Tearing effect line mode (POR = 0)
0 The tearing effect output line consists of V-blanking information only.
1 The tearing effect output line consists of both V-blanking and H-blanking information by set_tear_scanline (0x44).

The TE signal shall be active low when the display panel is in Sleep mode.

9.24 set_address_mode

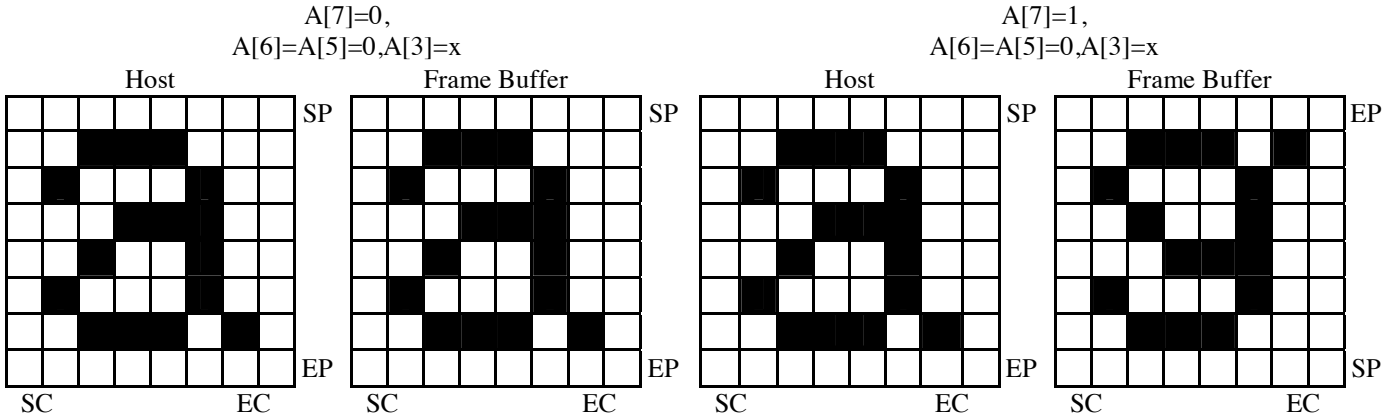
Command 0x36
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	0	1	1	0	36
Parameter 1	1	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	xx

Description
Set the read order from host processor to frame buffer by A[7:5] and A[3] and from frame buffer to the display panel by A[2:0] and A[4].

A[7] : Page address order (POR = 0)
This bit controls the order that pages of data are transferred from the host processor to the SSD1963's frame buffer.
0 Top to bottom, pages transferred from SP (Start Page) to EP (End Page).
1 Bottom to top, pages transferred from EP (End Page) to SP (Start Page).

Figure 9-11: A[7] Page Address Order



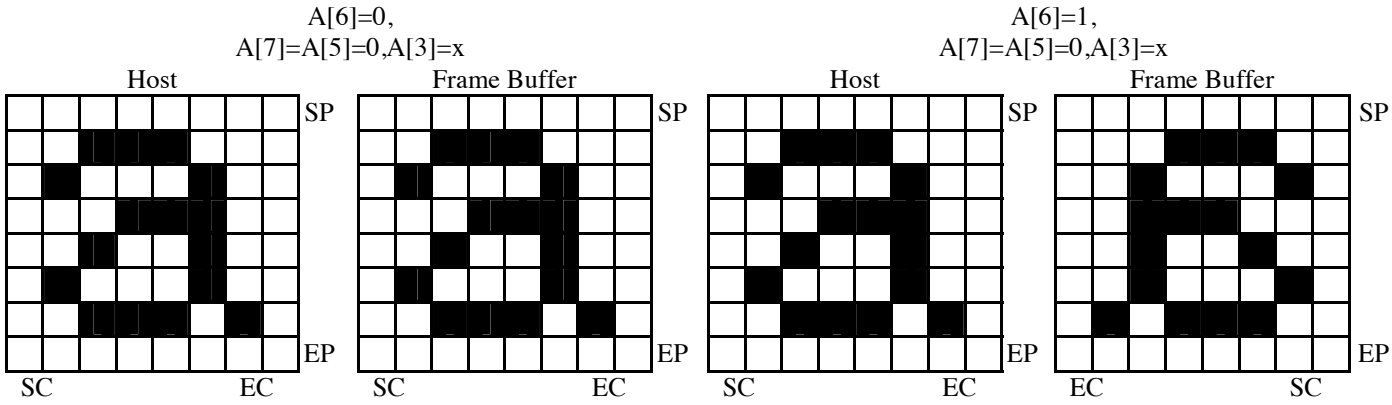
A[6] : Column address order (POR = 0)

This bit controls the order that columns of data are transferred from the host processor to the SSD1963's frame buffer.

0 Left to right, columns transferred from SC (Start Column) to EC (End Column).

1 Right to left, columns transferred from EC (End Column) to SC (Start Column).

Figure 9-12: A[6] Column Address Order



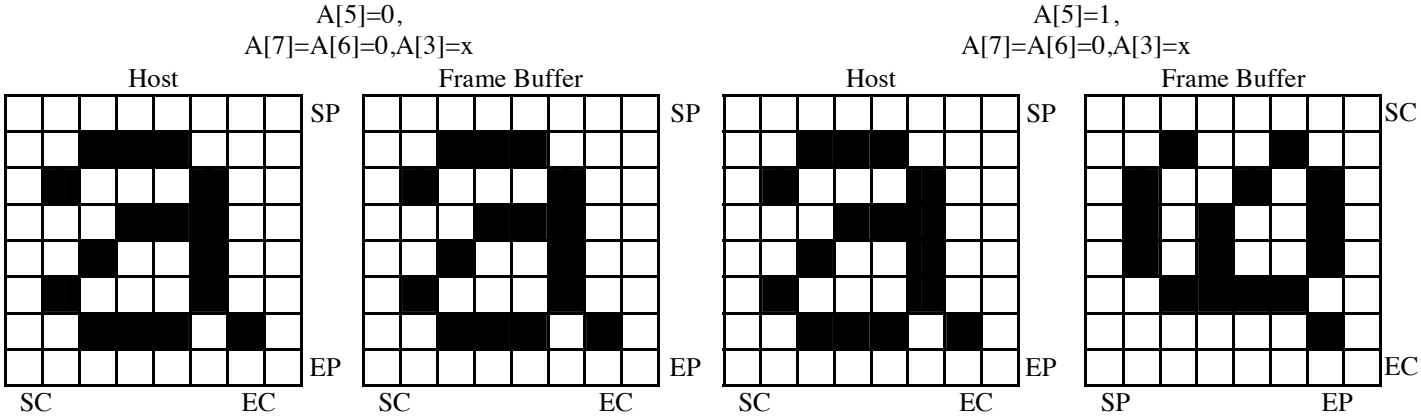
A[5] : Page / Column order (POR = 0)

This bit controls the order that columns of data are transferred from the host processor to the SSD1963's frame buffer.

0 Normal mode

1 Reverse mode

Figure 9-13: A[5] Page / Column Address Order



A[4] : Line address order (POR = 0)
This bit controls the display panel's horizontal line refresh order. The image shown on the display panel is unaffected, regardless of the bit setting.

0 LCD refresh from top line to bottom line.
1 LCD refresh from bottom line to top line.

A[3] : RGB / BGR order (POR = 0)
This bit controls the RGB data order transferred from the SSD1963's frame buffer to the display panel.

0 RGB
1 BGR

Figure 9-14: A[3] RGB Order



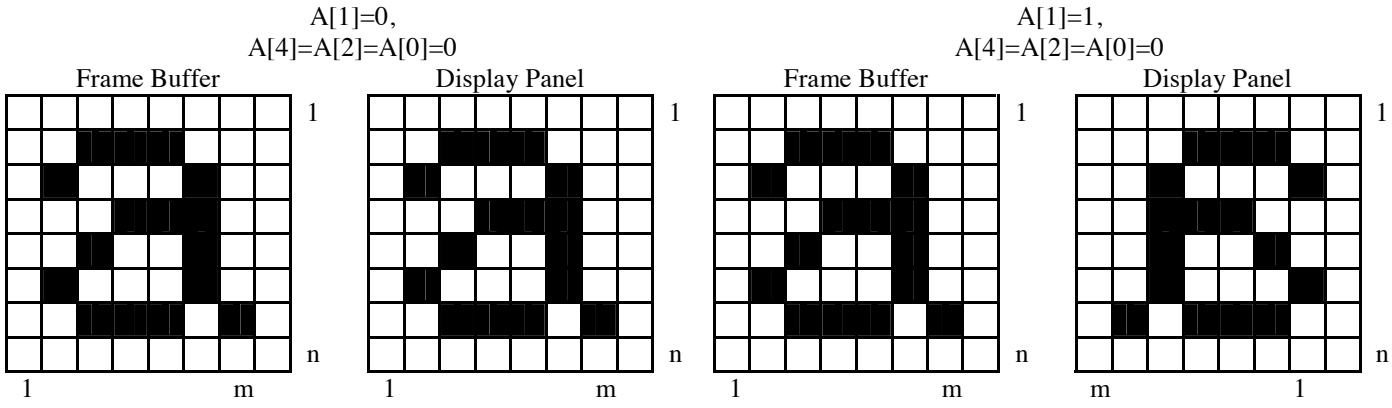
A[2] : Display data latch data (POR = 0)
This bit controls the display panel's vertical line data latch order. The image shown on the display panel is unaffected, regardless of the bit setting.

0 LCD refresh from left side to right side
1 LCD refresh from right side to left side

A[1] : Flip Horizontal (POR = 0)
This bit flips the image shown on the display panel left to right. No change is made to the frame buffer.

0 Normal
1 Flipped

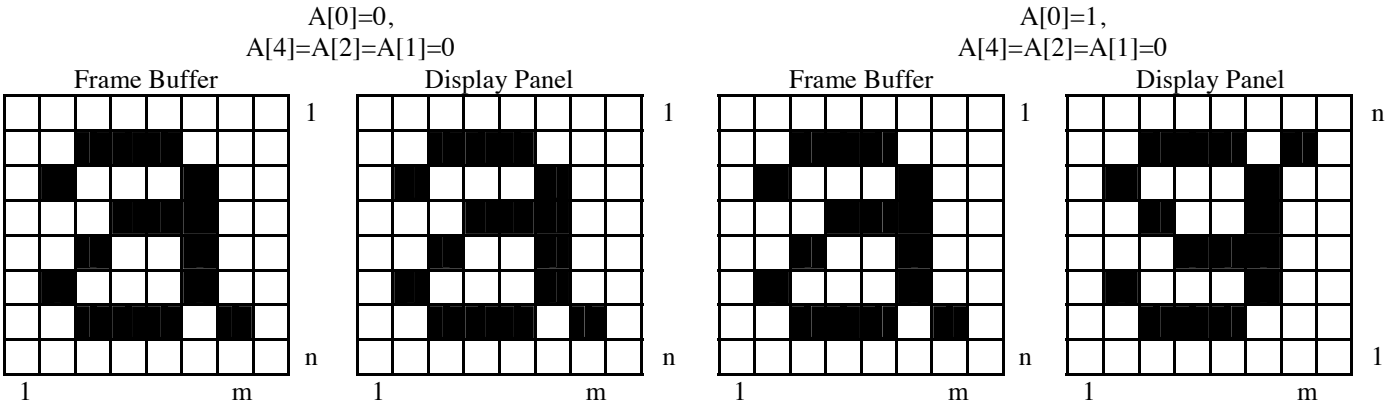
Figure 9-15: A[1] Flip Horizontal



A[0] : Flip Vertical (POR = 0)
This bit flips the image shown on the display panel top to bottom. No change is made to the frame buffer.

0	Normal
1	Flipped

Figure 9-16: A[0] Flip Vertical



9.25 set_scroll_start

Command 0x37
Parameters 2

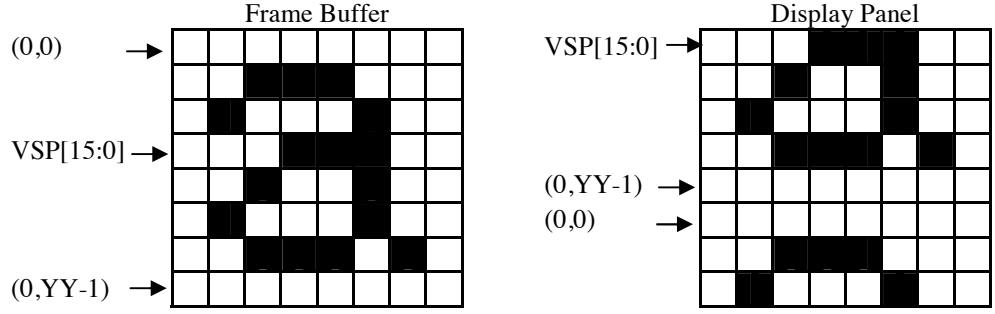
	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	0	1	1	0	1	1	37
Parameter 1	1	VSP_{15}	VSP_{14}	VSP_{13}	VSP_{12}	VSP_{11}	VSP_{10}	VSP_9	VSP_8	xx
Parameter 2	1	VSP_7	VSP_6	VSP_5	VSP_4	VSP_3	VSP_2	VSP_1	VSP_0	xx

Description
This command sets the start of the vertical scrolling area in the frame buffer. The vertical scrolling area is fully defined when this command is used with the set_scroll_area (0x33).

VSP[15:8] : High byte of the line number in frame buffer that is written to the display as the first line of the vertical scrolling area (POR = 00000000)
VSP[7:0] : Low byte of the line number in frame buffer that is written to the display as the first line of the vertical scrolling area (POR = 00000000)

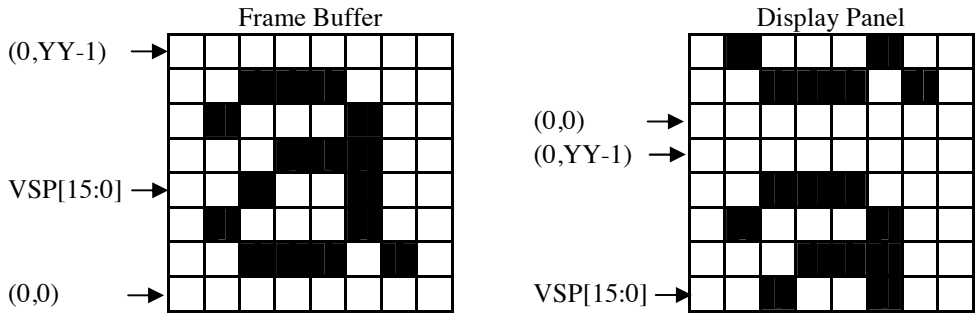
If set_address_mode (0x36) A[4] = 0:
Example:
When Top Fixed Area = Bottom Fixed Area = 0, Vertical Scrolling Area = YY and VSP = 3.

Figure 9-17: Set Scroll Start with set_address_mode (0x36) A[4] = 0



If set_address_mode (0x36) A[4] = 1:
Example:
When Top Fixed Area = Bottom Fixed Area = 0, Vertical Scrolling Area = YY and VSP = 3.

Figure 9-18: Set Scroll Start with set_address_mode (0x36) A[4] = 1



Note :
If set_address_mode, (0x36) A[4] = 0, TFA[15:0] - 1< VSP[15:0] < # of lines in frame buffer - BFA[15:0]
If set_address_mode, (0x36) A[4] = 1, BFA[15:0] - 1 < VSP[15:0] < # of lines in frame buffer - TFA[15:0]

9.26 exit_idle_mode

Command 0x38
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	1	0	0	0	38

Description
This command causes the SSD1963 to exit Idle Mode.
Full color depth is used for the display panel.

9.27 enter_idle_mode

Command 0x39
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	1	0	0	1	39

Description
This command causes the SSD1963 to enter Idle Mode.
In Idle Mode, color depth is reduced. Colors are shown on the display panel using the MSB of each of the R, G and B color components in the frame buffer.

Table 9-1 Enter Idle Mode memory content vs display color

Color	R_7 R_6 R_5 R_4 R_3 R_2 R_1 R_0	G_7 G_6 G_5 G_4 G_3 G_2 G_1 G_0	B_7 B_6 B_5 B_4 B_3 B_2 B_1 B_0
Black	0XXXXXXX	0XXXXXXX	0XXXXXXX
Blue	0XXXXXXX	0XXXXXXX	1XXXXXXX
Red	1XXXXXXX	0XXXXXXX	0XXXXXXX
Magenta	1XXXXXXX	0XXXXXXX	1XXXXXXX
Green	0XXXXXXX	1XXXXXXX	0XXXXXXX
Cyan	0XXXXXXX	1XXXXXXX	1XXXXXXX
Yellow	1XXXXXXX	1XXXXXXX	0XXXXXXX
White	1XXXXXXX	1XXXXXXX	1XXXXXXX

9.28 write_memory_continue

Command 0x3C
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	1	1	0	0	3C

Description
Transfer image information from the host processor interface to the SSD1963 from the last write_memory_continue (0x3C) or write_memory_start (0x2C).

If set_address_mode (0x36) A[5] = 0:

Data is written continuing from the pixel location after the write range of the previous write_memory_start (0x2C) or write_memory_continue (0x3C). The column address is then incremented and pixels are written to the frame buffer until the column address equals the End Column (EC) value. The column address is then reset to SC and the page address is incremented. Pixels are written to the frame buffer until the page address equals the End Page (EP) value and the column address equals the EC value, or the host processor sends another command. If the number of pixels exceeds (EC – SC + 1) * (EP – SP + 1) the extra pixels are ignored.

If set_address_mode (0x36) A[5] = 1:

Data is written continuing from the pixel location after the write range of the previous write_memory_start (0x2C) or write_memory_continue (0x3C). The page address is then incremented and pixels are written to the frame buffer until the page address equals the End Page (EP) value. The page address is then reset to SP and the column address is incremented. Pixels are written to the frame buffer until the column register equals the End column (EC) value and the page address equals the EP value, or the host processor sends another command. If the number of pixels exceeds (EC – SC + 1) * (EP – SP + 1) the extra pixels are ignored.

9.29 read_memory_continue

Command 0x3E
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	0	1	1	1	1	1	0	3E

Description
Read image data from the SSD1963 to host processor continuing after the last read_memory_continue (0x3E) or read_memory_start (0x2E).

If set_address_mode (0x36) A[5] = 0:

Pixels are read continuing from the pixel location after the read range of the previous read_memory_start (0x2E) or read_memory_continue (0x3E). The column address is then incremented and pixels are read from the frame buffer until the column address equals the End Column (EC) value. The column address is then reset to SC and the page address is incremented. Pixels are read from the frame buffer until the page address equals the End Page (EP) value and the column address equals the EC value, or the host processor sends another command.

If set_address_mode (0x36) A[5] = 1:

Pixels are read continuing from the pixel location after the read range of the previous read_memory_start (0x2E) or read_memory_continue (0x3E). The page address is then incremented and pixels are read from the frame buffer until the page address equals the End Page (EP) value. The page address is then reset to SP and the column address is incremented. Pixels are read from the frame buffer until the column address equals the End Column (EC) value and the page address equals the EP value, or the host processor sends another command.

9.30 set_tear_scanline

Command 0x44
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	1	0	0	0	1	0	0	44
Parameter 1	1	N_{15}	N_{14}	N_{13}	N_{12}	N_{11}	N_{10}	N_9	N_8	xx
Parameter 2	1	N_7	N_6	N_5	N_4	N_3	N_2	N_1	N_0	xx

Description
TE signal is sent from the SSD1963 to the host processor when the display panel refresh reaches the provided scanline, N.

N[15:8] : High byte of the scanline (POR = 00000000)
N[7:0] : Low byte of the scanline (POR = 00000000)

Note :

Set Tear Scanline with N = 0 is equivalent to set_tear_on (0x35) A[0] = 0.

This command takes affect on the frame following the current frame. Therefore, if the Tear Effect (TE) signal is already ON, the TE output shall continue to operate as programmed by the previous set_tear_on (0x35) or set_tear_scanline (0x44) until the end of the frame.

9.31 get_scanline

Command 0x45
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	0	1	0	0	0	1	0	1	45
Parameter 1	1	N_{15}	N_{14}	N_{13}	N_{12}	N_{11}	N_{10}	N_9	N_8	xx
Parameter 2	1	N_7	N_6	N_5	N_4	N_3	N_2	N_1	N_0	xx

Description
Get the current scan line, N.

N[15:8] : High byte of the current scanline (POR = 00000000)
N[7:0] : Low byte of the current scanline (POR = 00000000)

9.32 read_ddb

Command 0xA1
Parameters 5

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	0	0	0	0	1	A1
Parameter 1	1	SSL_{15}	SSL_{14}	SSL_{13}	SSL_{12}	SSL_{11}	SSL_{10}	SSL_9	SSL_8	xx
Parameter 2	1	SSL_7	SSL_6	SSL_5	SSL_4	SSL_3	SSL_2	SSL_1	SSL_0	xx
Parameter 3	1	$PROD_7$	$PROD_6$	$PROD_5$	$PROD_4$	$PROD_3$	$PROD_2$	$PROD_1$	$PROD_0$	xx
Parameter 4	1	0	0	0	0	0	REV_2	REV_1	REV_0	xx
Parameter 5	1	1	1	1	1	1	1	1	1	FF

Description
Read the DDB (Device Descriptor Block) information of SSD1963.

SSL[15:8] : Supplier ID of Solomon Systech Limited high byte, always 01h (POR = 00000001)
SSL[7:0] : Supplier ID of Solomon Systech Limited low byte, always 57h (POR = 010101110)
PROD[7:0] : Product ID, always 61h (POR = 01100001)
REV[2:0] : Revision code, always 01h (POR = 001)
Exit code, always FFh (POR = 11111111)

9.33 set_lcd_mode

Command 0xB0
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	0	0	0	0	B0
Parameter 1	1	0	0	A_5	A_4	A_3	A_2	A_1	A_0	xx
Parameter 2	1	0	B_6	B_5	0	0	0	0	0	xx
Parameter 3	1	0	0	0	0	0	HDP_{10}	HDP_9	HDP_8	xx
Parameter 4	1	HDP_7	HDP_6	HDP_5	HDP_4	HDP_3	HDP_2	HDP_1	HDP_0	xx
Parameter 5	1	0	0	0	0	0	VDP_{10}	VDP_9	VDP_8	xx
Parameter 6	1	VDP_7	VDP_6	VDP_5	VDP_4	VDP_3	VDP_2	VDP_1	VDP_0	xx
Parameter 7	1	0	0	G_5	G_4	G_3	G_2	G_1	G_0	xx

Description
Set the LCD panel mode and resolution
A[5] : TFT panel data width (POR = 0)
0 18-bit
1 24-bit

A[4] : TFT color depth enhancement enable (POR = 0)
0 Disable FRC or dithering
1 Enable FRC or dithering for color depth enhancement
If the panel data width was set to 24-bit, FRC and dithering feature will be disabled automatic regardless the value of this register.

A[3] : TFT FRC enable (POR = 0)
0 TFT dithering enable
1 TFT FRC enable

A[5]	A[4]	A[3]	TFT FRC	TFT dithering
0	0	X	Disable	Disable
0	1	0	Disable	Enable
0	1	1	Enable	Disable
1	X	X	Disable	Disable

A[2] : LSHIFT polarity (POR = 0)
Set the dot clock pulse polarity.

0 Data latch in falling edge
1 Data latch in rising edge

A[1] : LLINE polarity (POR = 0)
Set the horizontal sync pulse polarity.

0 Active low
1 Active high

A[0] : LFRAME polarity (POR = 0)
Set the vertical sync pulse polarity.

0 Active low
1 Active high

B[6:5] : TFT type (POR = 01)

00, 01 TFT mode
10 Serial RGB mode
11 Serial RGB+dummy mode

HDP [10:8] : High byte of the horizontal panel size (POR = 010)
HDP [7:0] : Low byte of the horizontal panel size (POR = 01111111)
Horizontal panel size = (HDP + 1) pixels

VDP [10:8] : High byte of the vertical panel size (POR = 001)
VDP [7:0] : Low byte of the vertical panel size (POR = 11011111)
Vertical panel size = (VDP + 1) lines

G[5:3] : Even line RGB sequence for serial TFT interface (POR = 000)

000 RGB
001 RBG
010 GRB
011 GBR
100 BRG
101 BGR
11x Reserved

G[2:0] : Odd line RGB sequence for serial TFT interface (POR = 000)

000 RGB
001 RBG
010 GRB
011 GBR
100 BRG
101 BGR
11x Reserved

9.34 get_lcd_mode

Command 0xB1
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	0	0	0	1	B1
Parameter 1	1	0	0	A_5	A_4	A_3	A_2	A_1	A_0	xx
Parameter 2	1	0	B_6	B_5	0	0	0	0	0	xx
Parameter 3	1	0	0	0	0	0	HDP_{10}	HDP_9	HDP_8	xx
Parameter 4	1	HDP_7	HDP_6	HDP_5	HDP_4	HDP_3	HDP_2	HDP_1	HDP_0	xx
Parameter 5	1	0	0	0	0	0	VDP_{10}	VDP_9	VDP_8	xx
Parameter 6	1	VDP_7	VDP_6	VDP_5	VDP_4	VDP_3	VDP_2	VDP_1	VDP_0	xx
Parameter 7	1	0	0	G_5	G_4	G_3	G_2	G_1	G_0	xx

Description
Get the current LCD panel mode and resolution

A[5] : TFT panel data width(POR = 0)
0 18-bit
1 24-bit

A[4] : TFT color depth enhancement enable(POR = 0)
0 Disable FRC or dithering
1 Enable FRC or dithering for color depth enhancement
If the panel data width was set to 24-bit, FRC and dithering feature will be disabled automatic regardless the value of this register.

A[3] : TFT FRC enable (POR = 0)
0 TFT dithering enable
1 TFT FRC enable

A[2] : LSHIFT polarity (POR = 0)
The dot clock pulse polarity.
0 Data latch in falling edge
1 Data latch in rising edge

A[1] : LLINE polarity (POR = 0)
The horizontal sync pulse polarity.
0 Active low
1 Active high

A[0] : LFRAME polarity (POR = 0)
The vertical sync pulse polarity.
0 Active low
1 Active high

B[6:5] : TFT type(POR = 01)
00, 01 TFT mode
10 Serial RGB mode
11 Serial RGB+dummy mode

HDP[10:8] : High byte of the horizontal panel size (POR = 010)
HDP[7:0] : Low byte of the horizontal panel size (POR = 01111111)

VDP[10:8] : High byte of the vertical panel size (POR = 001)

VDP[7:0] : Low byte of the vertical panel size (POR = 11011111)

G[5:3] : Even line RGB sequence (POR = 000)

000	RGB
001	RBG
010	GRB
011	GBR
100	BRG
101	BGR
11x	Reserved

G[2:0] : Odd line RGB sequence (POR = 000)

000	RGB
001	RBG
010	GRB
011	GBR
100	BRG
101	BGR
11x	Reserved

9.35 set_hori_period

Command 0xB4
Parameters 8

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	0	1	0	0	B4
Parameter 1	1	0	0	0	0	0	HT_{10}	HT_9	HT_8	xx
Parameter 2	1	HT_7	HT_6	HT_5	HT_4	HT_3	HT_2	HT_1	HT_0	xx
Parameter 3	1	0	0	0	0	0	HPS_{10}	HPS_9	HPS_8	xx
Parameter 4	1	HPS_7	HPS_6	HPS_5	HPS_4	HPS_3	HPS_2	HPS_1	HPS_0	xx
Parameter 5	1	0	HPW_6	HPW_5	HPW_4	HPW_3	HPW_2	HPW_1	HPW_0	xx
Parameter 6	1	0	0	0	0	0	LPS_{10}	LPS_9	LPS_8	xx
Parameter 7	1	LPS_7	LPS_6	LPS_5	LPS_4	LPS_3	LPS_2	LPS_1	LPS_0	xx
Parameter 8	1	0	0	0	0	0	0	$LPSPP_1$	$LPSPP_0$	xx

Description
Set front porch and back porch

HT[10:8] : High byte of horizontal total period (display + non-display) in pixel clock (POR = 010)
HT[7:0] : Low byte of the horizontal total period (display + non-display) in pixel clock (POR = 10101111)
Horizontal total period = (HT + 1) pixels

HPS[10:8] : High byte of the non-display period between the start of the horizontal sync (LLINE) signal and the first display data. (POR = 000)
HPS[7:0] : Low byte of the non-display period between the start of the horizontal sync (LLINE) signal and the first display data. (POR = 00100000)
For TFT : Horizontal Sync Pulse Start Position = HPS pixels
For Serial TFT : Horizontal Sync Pulse Start Position = HPS pixels + LPSPP subpixels

HPW[6:0] : Set the horizontal sync pulse width (LLINE) in pixel clock. (POR = 0000111)
Horizontal Sync Pulse Width = (HPW + 1) pixels

LPS[10:8] : Set the horizontal sync pulse (LLINE) start location in pixel clock. (POR = 000)
LPS[7:0] : Set the horizontal sync pulse width (LLINE) in start. (POR = 00000000)
Horizontal Display Period Start Position = LPS pixels

LPSPP[1:0] : Set the horizontal sync pulse subpixel start position for serial TFT interface (POR = 00)

Timing refer to Figure 13-5.

9.36 get_hori_period

Command 0xB5
Parameters 8

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	0	1	0	1	B5
Parameter 1	1	0	0	0	0	0	HT_{10}	HT_9	HT_8	xx
Parameter 2	1	HT_7	HT_6	HT_5	HT_4	HT_3	HT_2	HT_1	HT_0	xx
Parameter 3	1	0	0	0	0	0	HPS_{10}	HPS_9	HPS_8	xx
Parameter 4	1	HPS_7	HPS_6	HPS_5	HPS_4	HPS_3	HPS_2	HPS_1	HPS_0	xx
Parameter 5	1	0	HPW_6	HPW_5	HPW_4	HPW_3	HPW_2	HPW_1	HPW_0	xx
Parameter 6	1	0	0	0	0	0	LPS_{10}	LPS_9	LPS_8	xx
Parameter 7	1	LPS_7	LPS_6	LPS_5	LPS_4	LPS_3	LPS_2	LPS_1	LPS_0	xx
Parameter 8	1	0	0	0	0	0	0	$LPSPP_1$	$LPSPP_0$	xx

Description
Get current front porch and back porch settings

HT[10:8] : High byte of the horizontal total period (display + non-display) in pixel clock (POR = 010)
HT[7:0] : Low byte of the horizontal total period (display + non-display) in pixel clock (POR = 10101111)

HPS[10:8] : High byte of the non-display period between the start of the horizontal sync (LLINE) signal and the first display data. (POR = 000)
HPS[7:0] : Low byte of the non-display period between the start of the horizontal sync (LLINE) signal and the first display data. (POR = 00100000)

HPW[6:0] : The horizontal sync pulse width (LLINE) in pixel clock. (POR = 0000111)

LPS[10:8] : High byte of the horizontal sync pulse (LLINE) start location in pixel clock. (POR = 000)
LPS[7:0] : Low byte of the horizontal sync pulse width (LLINE) in start. (POR = 00000000)

LPSPP[1:0] : The horizontal sync pulse subpixel start position (POR = 00)

9.37 set_vert_period

Command 0xB6
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	0	1	1	0	B6
Parameter 1	1	0	0	0	0	0	VT_{10}	VT_9	VT_8	xx
Parameter 2	1	VT_7	VT_6	VT_5	VT_4	VT_3	VT_2	VT_1	VT_0	xx
Parameter 3	1	0	0	0	0	0	VPS_{10}	VPS_9	VPS_8	xx
Parameter 4	1	VPS_7	VPS_6	VPS_5	VPS_4	VPS_3	VPS_2	VPS_1	VPS_0	xx
Parameter 5	1	0	VPW_6	VPW_5	VPW_4	VPW_3	VPW_2	VPW_1	VPW_0	xx
Parameter 6	1	0	0	0	0	0	FPS_{10}	FPS_9	FPS_8	xx
Parameter 7	1	FPS_7	FPS_6	FPS_5	FPS_4	FPS_3	FPS_2	FPS_1	FPS_0	xx

Description
Set the vertical blanking interval between last scan line and next LFRAME pulse

VT[10:8] : High byte of the vertical total (display + non-display) period in lines (POR = 001)
VT[7:0] : Low byte of the vertical total (display + non-display) period in lines (POR = 11101111)
Vertical Total = (VT + 1) lines

VPS[10:8] : High byte the non-display period in lines between the start of the frame and the first display data in line. (POR = 000)
VPS[7:0] : The non-display period in lines between the start of the frame and the first display data in line. (POR = 00000100)
Vertical Sync Pulse Start Position = VPS lines

VPW[6:0] : Set the vertical sync pulse width (LFRAME) in lines. (POR = 000001)
Vertical Sync Pulse Width = (VPW + 1) lines

FPS[10:8] : High byte of the vertical sync pulse (LFRAME) start location in lines. (POR = 000)
FPS[7:0] : Low byte of the vertical sync pulse (LFRAME) start location in lines. (POR = 00000000)
Vertical Display Period Start Position = FPS lines

Timing refer to Figure 13-5.

9.38 get_vert_period

Command 0xB7
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	0	1	1	1	B7
Parameter 1	1	0	0	0	0	0	VT_{10}	VT_9	VT_8	xx
Parameter 2	1	VT_7	VT_6	VT_5	VT_4	VT_3	VT_2	VT_1	VT_0	xx
Parameter 3	1	0	0	0	0	0	VPS_{10}	VPS_9	VPS_8	xx
Parameter 4	1	VPS_7	VPS_6	VPS_5	VPS_4	VPS_3	VPS_2	VPS_1	VPS_0	xx
Parameter 5	1	0	VPW_6	VPW_5	VPW_4	VPW_3	VPW_2	VPW_1	VPW_0	xx
Parameter 6	1	0	0	0	0	0	FPS_{10}	FPS_9	FPS_8	xx
Parameter 7	1	FPS_7	FPS_6	FPS_5	FPS_4	FPS_3	FPS_2	FPS_1	FPS_0	xx

Description
Get the vertical blanking interval between last scan line and next LFRAME pulse

VT[10:8] : High byte of the vertical total (display + non-display) period in lines (POR = 001)
VT[7:0] : Low byte of the vertical total (display + non-display) period in lines (POR = 01111111)

VPS[10:8] : High byte of the non-display period in lines between the start of the frame and the first display data in line. (POR = 000)
VPS[7:0] : Low byte of the non-display period in lines between the start of the frame and the first display data in line. (POR = 00000100)

VPW[6:0] : The vertical sync pulse width (LFRAME) in lines. (POR = 000001)

FPS[10:8] : High byte of the vertical sync pulse (LFRAME) start location in lines. (POR = 000)
FPS[7:0] : Low byte of the vertical sync pulse (LFRAME) start location in lines. (POR = 00000000)

9.39 set_gpio_conf

Command 0xB8
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	0	0	0	B8
Parameter 1	1	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	xx
Parameter 2	1	0	0	0	0	0	0	0	B_0	xx

Description
Set the GPIOs configuration. If the GPIOs are not used for LCD, set the direction. Otherwise, they are toggled with LCD signals by 0xC0 – 0xCF.

A[7] : GPIO3 configuration (POR = 0)
0 GPIO3 is controlled by host
1 GPIO3 is controlled by LCDC

A[6] : GPIO2 configuration (POR = 0)
0 GPIO2 is controlled by host
1 GPIO2 is controlled by LCDC

A[5] : GPIO1 configuration (POR = 0)
0 GPIO1 is controlled by host
1 GPIO1 is controlled by LCDC

A[4] : GPIO0 configuration (POR = 0)
0 GPIO0 is controlled by host
1 GPIO0 is controlled by LCDC

A[3] : GPIO3 direction (POR = 0)
0 GPIO3 is input
1 GPIO3 is output

A[2] : GPIO3 direction (POR = 0)
0 GPIO2 is input
1 GPIO2 is output

A[1] : GPIO1 direction (POR = 0)
0 GPIO1 is input
1 GPIO1 is output

A[0] : GPIO0 direction (POR = 0)
0 GPIO0 is input
1 GPIO0 is output

B[0] : GPIO0 direction (POR = 0)
0 GPIO0 is used to control the panel power with enter_sleep_mode (0x10) or exit_sleep_mode (0x11).
1 GPIO0 is used as normal GPIO

9.40 get_gpio_conf

Command 0xB9
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	0	0	1	B9
Parameter 1	1	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	xx
Parameter 2	1	0	0	0	0	0	0	0	B_0	xx

Description
Get the current GPIOs configuration

A[7] : GPIO3 configuration (POR = 0)
0 GPIO3 is controlled by host
1 GPIO3 is controlled by LCDC

A[6] : GPIO2 configuration (POR = 0)
0 GPIO2 is controlled by host
1 GPIO2 is controlled by LCDC

A[5] : GPIO1 configuration (POR = 0)
0 GPIO1 is controlled by host
1 GPIO1 is controlled by LCDC

A[4] : GPIO0 configuration (POR = 0)
0 GPIO0 is controlled by host
1 GPIO0 is controlled by LCDC

A[3] : GPIO3 direction (POR = 0)
0 GPIO3 is input
1 GPIO3 is output

A[2] : GPIO3 direction (POR = 0)
0 GPIO2 is input
1 GPIO2 is output

A[1] : GPIO1 direction (POR = 0)
0 GPIO1 is input
1 GPIO1 is output

A[0] : GPIO0 direction (POR = 0)
0 GPIO0 is input
1 GPIO0 is output

B[0] : GPIO0 direction (POR = 0)
0 GPIO0 is used to control the panel power with enter_sleep_mode (0x10) or exit_sleep_mode (0x11)
1 GPIO0 is used as normal GPIO

9.41 set_gpio_value

Command 0xBA
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	0	1	0	BA
Parameter 1	1	0	0	0	0	A_3	A_2	A_1	A_0	xx

Description
Set GPIO value for GPIO configured as output

A[3] : GPIO3 value (POR = 0)
0 GPIO3 outputs 0
1 GPIO3 outputs 1

A[2] : GPIO2 value (POR = 0)
0 GPIO2 outputs 0
1 GPIO2 outputs 1

A[1] : GPIO1 value (POR = 0)
0 GPIO1 outputs 0
1 GPIO1 outputs 1

A[0] : GPIO0 value (POR = 0)
0 GPIO0 outputs 0
1 GPIO0 outputs 1

9.42 get_gpio_status

Command 0xBB
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	0	1	1	BB
Parameter 1	1	0	0	0	0	A_3	A_2	A_1	A_0	xx

Description
Read current GPIO status. If the individual GPIO was configured as input, the value is the status of the corresponding pin. Otherwise, it is the programmed value.

A[3] : GPIO3 value (POR : depends on pad value)
0 GPIO3 is pulled low
1 GPIO3 is pulled high

A[2] : GPIO2 value (POR : depends on pad value)
0 GPIO2 is pulled low
1 GPIO2 is pulled high

A[1] : GPIO1 value (POR : depends on pad value)
0 GPIO1 is pulled low
1 GPIO1 is pulled high

A[0] : GPIO0 value (POR : depends on pad value)
0 GPIO0 is pulled low
1 GPIO0 is pulled high

9.43 set_post_proc

Command 0xBC
Parameters 4

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	1	0	0	BC
Parameter 1	1	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx
Parameter 3	1	C_7	C_6	C_5	C_4	C_3	C_2	C_1	C_0	xx
Parameter 4	1	0	0	0	0	0	0	0	D_0	xx

Description
Set the image post processor

A[7:0] : Set the contrast value (POR = 01000000)

B[7:0] : Set the brightness value (POR = 10000000)

C[7:0] : Set the saturation value (POR = 01000000)

D[0] : Post Processor Enable (POR = 0)
0 Disable the postprocessor
1 Enable the postprocessor

9.44 get_post_proc

Command 0xBD
Parameters 4

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	1	0	1	BD
Parameter 1	1	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx
Parameter 3	1	C_7	C_6	C_5	C_4	C_3	C_2	C_1	C_0	xx
Parameter 4	1	0	0	0	0	0	0	0	D_0	xx

Description
Get the image post processor

A[7:0] : Get the contrast value (POR = 01000000)

B[7:0] : Get the brightness value (POR = 10000000)

C[7:0] : Get the saturation value (POR = 01000000)

D[0] : Post Processor Enable (POR = 0)
0 Disable the postprocessor
1 Enable the postprocessor

9.45 set_pwm_conf

Command 0xBE
Parameters 6

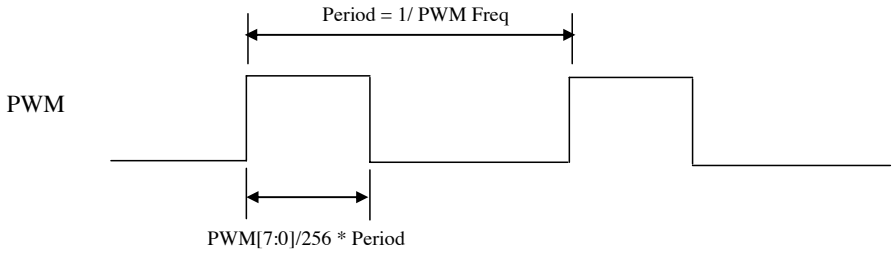
	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	1	1	0	BE
Parameter 1	1	$PWMF_7$	$PWMF_6$	$PWMF_5$	$PWMF_4$	$PWMF_3$	$PWMF_2$	$PWMF_1$	$PWMF_0$	xx
Parameter 2	1	PWM_7	PWM_6	PWM_5	PWM_4	PWM_3	PWM_2	PWM_1	PWM_0	xx
Parameter 3	1	0	0	0	0	C_3	0	0	C_0	xx
Parameter 4	1	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	xx
Parameter 5	1	E_7	E_6	E_5	E_4	E_3	E_2	E_1	E_0	xx
Parameter 6	1	0	0	0	0	F_3	F_2	F_1	F_0	xx

Description
Set the PWM configuration

PWMF[7:0] : Set the PWM frequency in system clock (POR = 00000000)
PWM signal frequency = PLL clock / (256 * PWMF[7:0]) / 256

PWM[7:0] : Set the PWM duty cycle (POR = 00000000)
PWM duty cycle = PWM[7:0] / 256 for DBC disable (0xD0] A0 = 0
If DBC enable (0xD0] A0 = 1, these parameter will be ignored
Note : PWM always 0 if PWM[7:0] = 00h

Figure 9-19: PWM signal



C[3] : PWM configuration (POR = 0)
0 PWM controlled by host
1 PWM controlled by DBC

C[0] : PWM enable (POR = 0)
0 PWM disable
1 PWM enable

D[7:0] : DBC manual brightness (POR = 00000000)
Set the manual brightness level. When Manual Brightness Mode (0xD0) A[6] is enabled, the final DBC duty cycle output will be multiplied by this value / 255.
PWM duty cycle = DBC output * D[7:0] / 255
00 Dimmest
FF brightest

E[7:0] : DBC minimum brightness (POR = 00000000)
Set the minimum brightness level. WhenManual Brightness Mode (0xD0) A[6] is enabled, DBC duty cycle output will be limited by this value. This will prevent from backlight being too dark or off.
00 Dimmest
FF Brightest

F[3:0] : Brightness prescaler (POR = 0000)
Set the brightness prescaler to control how gradually the manual brightness is changed between different levels. There is a filter will undergo a number of iterations before the manual brightness saturated. This parameter is valid when Transition Effect enable (0xD0) A5 = 1
The iteration ration = system frequency / Divcode / 32768

F[3:0]	Divcode
0000	off
0001	1
0010	2
0011	3
0100	4
0101	6
0110	8
0111	12
1000	16
1001	24
1010	32
1011	48
1100	64
1101	96
1110	128
1111	192

9.46 get_pwm_conf

Command 0xBF
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	0	1	1	1	1	1	1	BF
Parameter 1	1	$PWMF_7$	$PWMF_6$	$PWMF_5$	$PWMF_4$	$PWMF_3$	$PWMF_2$	$PWMF_1$	$PWMF_0$	xx
Parameter 2	1	PWM_7	PWM_6	PWM_5	PWM_4	PWM_3	PWM_2	PWM_1	PWM_0	xx
Parameter 3	1	0	0	0	0	C_3	0	0	C_0	xx
Parameter 4	1	D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	xx
Parameter 5	1	E_7	E_6	E_5	E_4	E_3	E_2	E_1	E_0	xx
Parameter 6	1	0	0	0	0	F_3	F_2	F_1	F_0	xx
Parameter 7	1	G_7	G_6	G_5	G_4	G_3	G_2	G_1	G_0	xx

Description
Get the PWM configuration

PWMF[7:0] : Get the PWM frequency in system clock (POR = 00000000)

PWM[7:0] : Get the PWM duty cycle (POR = 00000000)

C[3] : PWM configuration (POR = 0)
0 PWM controlled by host
1 PWM controlled by DBC

C[0] : PWM enable (POR = 0)
0 PWM disable
1 PWM enable

D[7:0] : DBC manual brightness (POR = 00000000)
Get the brightness level
00 Dimmest
FF brightest

E[7:0] : DBC minimum brightness (POR = 00000000)
Get the minimum brightness level.

00	Dimmest
FF	Brightest

F[3:0] : Brightness prescaler (POR = 0000)
Get the brightness prescaler

G[7:0] : Dynamic backlight duty cycle : Get the current PWM duty cycle controlled by PWM (POR = 00000000)

9.47 set_lcd_gen0

Command 0xC0
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	0	0	0	C0
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF0_{10}$	$GF0_9$	$GF0_8$	xx
Parameter 3	1	$GF0_7$	$GF0_6$	$GF0_5$	$GF0_4$	$GF0_3$	$GF0_2$	$GF0_1$	$GF0_0$	xx
Parameter 4	1	0	0	0	0	0	$GR0_{10}$	$GR0_9$	$GR0_8$	xx
Parameter 5	1	$GR0_7$	$GR0_6$	$GR0_5$	$GR0_4$	$GR0_3$	$GR0_2$	$GR0_1$	$GR0_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP0_{10}$	$GP0_9$	$GP0_8$	xx
Parameter 7	1	$GP0_7$	$GP0_6$	$GP0_5$	$GP0_4$	$GP0_3$	$GP0_2$	$GP0_1$	$GP0_0$	xx

Description
Set the rise, fall, period and toggling properties of LCD signal generator 0

A[7] : Reset LCD generator 0 at every frame start
- 0 The generator 0 will not reset in the starting point of a frame
- 1 The generator 0 will reset in the starting point of a frame

GF0[10:8] : The highest 3 bits of the generator 0 falling position (POR = 000)
GF0[7:0] : The lower byte of the generator 0 falling position (POR = 00000001)

GR0[10:8] : The highest 3 bits of the generator 0 rising position (POR = 000)
GR0[7:0] : The lower byte of the generator 0 rising position (POR = 00000000)

F[7] : Force the generator 0 output to 0 in non-display period
- 0 generator 0 is normal
- 1 generator 0 output is forced to 0 in non-display period

F[6:5] : Force the generator 0 output to 0 in odd or even lines
- 00 generator 0 is normal in both odd and even lines
- 01 generator 0 output is force to 0 in odd lines
- 10 generator 0 output is force to 0 in even lines
- 11 generator 0 is normal in both odd and even line

F[4:3] : Generator 0 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP0[10:8] : The highest 3 bits of the generator 0 period (POR = 100)
GP0[7:0] : The lower byte of the generator 0 period (POR = 00000000)

9.48 get_lcd_gen0

Command 0xC1
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	0	0	1	C1
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF0_{10}$	$GF0_9$	$GF0_8$	xx
Parameter 3	1	$GF0_7$	$GF0_6$	$GF0_5$	$GF0_4$	$GF0_3$	$GF0_2$	$GF0_1$	$GF0_0$	xx
Parameter 4	1	0	0	0	0	0	$GR0_{10}$	$GR0_9$	$GR0_8$	xx
Parameter 5	1	$GR0_7$	$GR0_6$	$GR0_5$	$GR0_4$	$GR0_3$	$GR0_2$	$GR0_1$	$GR0_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP0_{10}$	$GP0_9$	$GP0_8$	xx
Parameter 7	1	$GP0_7$	$GP0_6$	$GP0_5$	$GP0_4$	$GP0_3$	$GP0_2$	$GP0_1$	$GP0_0$	xx

Description
Get the rise, fall, period and toggling properties of LCD signal generator 0

A[7] : Reset LCD generator 0 at every frame start
- 0 The generator 0 will not reset in the starting point of a frame
- 1 The generator 0 will reset in the starting point of a frame

GF0[10:8] : The highest 3 bits of the generator 0 falling position (POR = 000)
GF0[7:0] : The lower byte of the generator 0 falling position (POR = 00000001)

GR0[10:8] : The highest 3 bits of the generator 0 rising position (POR = 000)
GR0[7:0] : The lower byte of the generator 0 rising position (POR = 00000000)

F[7] : Force the generator 0 output to 0 in non-display period
- 0 generator 0 is normal
- 1 generator 0 output is forced to 0 in non-display period

F[6:5] : Force the generator 0 output to 0 in odd or even lines
- 00 generator 0 is normal in both odd and even lines
- 01 generator 0 output is force to 0 in odd lines
- 10 generator 0 output is force to 0 in even lines
- 11 generator 0 is normal in both odd and even line

F[4:3] : Generator 0 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP0[10:8] : The highest 3 bits of the generator 0 period (POR = 100)
GP0[7:0] : The lower byte of the generator 0 period (POR = 00000000)

9.49 set_lcd_gen1

Command 0xC2
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	0	1	0	C2
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF1_{10}$	$GF1_9$	$GF1_8$	xx
Parameter 3	1	$GF1_7$	$GF1_6$	$GF1_5$	$GF1_4$	$GF1_3$	$GF1_2$	$GF1_1$	$GF1_0$	xx
Parameter 4	1	0	0	0	0	0	$GR1_{10}$	$GR1_9$	$GR1_8$	xx
Parameter 5	1	$GR1_7$	$GR1_6$	$GR1_5$	$GR1_4$	$GR1_3$	$GR1_2$	$GR1_1$	$GR1_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP1_{10}$	$GP1_9$	$GP1_8$	xx
Parameter 7	1	$GP1_7$	$GP1_6$	$GP1_5$	$GP1_4$	$GP1_3$	$GP1_2$	$GP1_1$	$GP1_0$	xx

Description
Set the rise, fall, period and toggling properties of LCD signal generator 1

A[7] : Reset LCD generator 1 at every frame start
- 0 The generator 1 will not reset in the starting point of a frame
- 1 The generator 1 will reset in the starting point of a frame

GF1[10:8] : The highest 3 bits of the generator 1 falling position (POR = 000)
GF1[7:0] : The lower byte of the generator 1 falling position (POR = 00000001)

GR1[10:8] : The highest 3 bits of the generator 1 rising position (POR = 000)
GR1[7:0] : The lower byte of the generator 1 rising position (POR = 00000000)

F[7] : Force the generator 1 output to 0 in non-display period
- 0 generator 1 is normal
- 1 generator 1 output is forced to 0 in non-display period

F[6:5] : Force the generator 1 output to 0 in odd or even lines
- 00 generator 1 is normal in both odd and even lines
- 01 generator 1 output is force to 0 in odd lines
- 10 generator 1 output is force to 0 in even lines
- 11 generator 1 is normal in both odd and even line

F[4:3] : Generator 1 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP1[10:8] : The highest 3 bits of the generator 1 period (POR = 100)
GP1[7:0] : The lower byte of the generator 1 period (POR = 00000000)

9.50 get_lcd_gen1

Command 0xC3
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	0	1	1	C3
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF1_{10}$	$GF1_9$	$GF1_8$	xx
Parameter 3	1	$GF1_7$	$GF1_6$	$GF1_5$	$GF1_4$	$GF1_3$	$GF1_2$	$GF1_1$	$GF1_0$	xx
Parameter 4	1	0	0	0	0	0	$GR1_{10}$	$GR1_9$	$GR1_8$	xx
Parameter 5	1	$GR1_7$	$GR1_6$	$GR1_5$	$GR1_4$	$GR1_3$	$GR1_2$	$GR1_1$	$GR1_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP1_{10}$	$GP1_9$	$GP1_8$	xx
Parameter 7	1	$GP1_7$	$GP1_6$	$GP1_5$	$GP1_4$	$GP1_3$	$GP1_2$	$GP1_1$	$GP1_0$	xx

Description
Get the rise, fall, period and toggling properties of LCD signal generator 1

A[7] : Reset LCD generator 1 at every frame start
- 0 The generator 1 will not reset in the starting point of a frame
- 1 The generator 1 will reset in the starting point of a frame

GF1[10:8] : The highest 3 bits of the generator 1 falling position (POR = 000)
GF1[7:0] : The lower byte of the generator 1 falling position (POR = 00000001)

GR1[10:8] : The highest 3 bits of the generator 1 rising position (POR = 000)
GR1[7:0] : The lower byte of the generator 1 rising position (POR = 00000000)

F[7] : Force the generator 1 output to 0 in non-display period
- 0 generator 1 is normal
- 1 generator 1 output is forced to 0 in non-display period

F[6:5] : Force the generator 1 output to 0 in odd or even lines
- 00 generator 1 is normal in both odd and even lines
- 01 generator 1 output is force to 0 in odd lines
- 10 generator 1 output is force to 0 in even lines
- 11 generator 1 is normal in both odd and even line

F[4:3] : Generator 1 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP1[10:8] : The highest 3 bits of the generator 1 period (POR = 100)
GP1[7:0] : The lower byte of the generator 1 period (POR = 00000000)

9.51 set_lcd_gen2

Command 0xC4
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	1	0	0	C4
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF2_{10}$	$GF2_9$	$GF2_8$	xx
Parameter 3	1	$GF2_7$	$GF2_6$	$GF2_5$	$GF2_4$	$GF2_3$	$GF2_2$	$GF2_1$	$GF2_0$	xx
Parameter 4	1	0	0	0	0	0	$GR2_{10}$	$GR2_9$	$GR2_8$	xx
Parameter 5	1	$GR2_7$	$GR2_6$	$GR2_5$	$GR2_4$	$GR2_3$	$GR2_2$	$GR2_1$	$GR2_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP2_{10}$	$GP2_9$	$GP2_8$	xx
Parameter 7	1	$GP2_7$	$GP2_6$	$GP2_5$	$GP2_4$	$GP2_3$	$GP2_2$	$GP2_1$	$GP2_0$	xx

Description
Set the rise, fall, period and toggling properties of LCD signal generator 2

A[7] : Reset LCD generator 2 at every frame start
- 0 The generator 2 will not reset in the starting point of a frame
- 1 The generator 2 will reset in the starting point of a frame

GF2[10:8] : The highest 3 bits of the generator 2 falling position (POR = 000)
GF2[7:0] : The lower byte of the generator 2 falling position (POR = 00000001)

GR2[10:8] : The highest 3 bits of the generator 2 rising position (POR = 000)
GR2[7:0] : The lower byte of the generator 2 rising position (POR = 00000000)

F[7] : Force the generator 2 output to 0 in non-display period
- 0 generator 2 is normal
- 1 generator 2 output is forced to 0 in non-display period

F[6:5] : Force the generator 2 output to 0 in odd or even lines
- 00 generator 2 is normal in both odd and even lines
- 01 generator 2 output is force to 0 in odd lines
- 10 generator 2 output is force to 0 in even lines
- 11 generator 2 is normal in both odd and even line

F[4:3] : Generator 2 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP2[10:8] : The highest 3 bits of the generator 2 period (POR = 100)
GP2[7:0] : The lower byte of the generator 2 period (POR = 00000000)

9.52 get_lcd_gen2

Command 0xC5
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	1	0	1	C5
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF2_{10}$	$GF2_9$	$GF2_8$	xx
Parameter 3	1	$GF2_7$	$GF2_6$	$GF2_5$	$GF2_4$	$GF2_3$	$GF2_2$	$GF2_1$	$GF2_0$	xx
Parameter 4	1	0	0	0	0	0	$GR2_{10}$	$GR2_9$	$GR2_8$	xx
Parameter 5	1	$GR2_7$	$GR2_6$	$GR2_5$	$GR2_4$	$GR2_3$	$GR2_2$	$GR2_1$	$GR2_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP2_{10}$	$GP2_9$	$GP2_8$	xx
Parameter 7	1	$GP2_7$	$GP2_6$	$GP2_5$	$GP2_4$	$GP2_3$	$GP2_2$	$GP2_1$	$GP2_0$	xx

Description
Get the rise, fall, period and toggling properties of LCD signal generator 2

A[7] : Reset LCD generator 2 at every frame start
- 0 The generator 2 will not reset in the starting point of a frame
- 1 The generator 2 will reset in the starting point of a frame

GF2[10:8] : The highest 3 bits of the generator 2 falling position (POR = 000)
GF2[7:0] : The lower byte of the generator 2 falling position (POR = 00000001)

GR2[10:8] : The highest 3 bits of the generator 2 rising position (POR = 000)
GR2[7:0] : The lower byte of the generator 2 rising position (POR = 00000000)

F[7] : Force the generator 2 output to 0 in non-display period
- 0 generator 2 is normal
- 1 generator 2 output is forced to 0 in non-display period

F[6:5] : Force the generator 2 output to 0 in odd or even lines
- 00 generator 2 is normal in both odd and even lines
- 01 generator 2 output is force to 0 in odd lines
- 10 generator 2 output is force to 0 in even lines
- 11 generator 2 is normal in both odd and even line

F[4:3] : Generator 2 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP2[10:8] : The highest 3 bits of the generator 2 period (POR = 100)
GP2[7:0] : The lower byte of the generator 2 period (POR = 00000000)

9.53 set_lcd_gen3

Command 0xC6
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	1	1	0	C6
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF3_{10}$	$GF3_9$	$GF3_8$	xx
Parameter 3	1	$GF3_7$	$GF3_6$	$GF3_5$	$GF3_4$	$GF3_3$	$GF3_2$	$GF3_1$	$GF3_0$	xx
Parameter 4	1	0	0	0	0	0	$GR3_{10}$	$GR3_9$	$GR3_8$	xx
Parameter 5	1	$GR3_7$	$GR3_6$	$GR3_5$	$GR3_4$	$GR3_3$	$GR3_2$	$GR3_1$	$GR3_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP3_{10}$	$GP3_9$	$GP3_8$	xx
Parameter 7	1	$GP3_7$	$GP3_6$	$GP3_5$	$GP3_4$	$GP3_3$	$GP3_2$	$GP3_1$	$GP3_0$	xx

Description
Set the rise, fall, period and toggling properties of LCD signal generator 3

A[7] : Reset LCD generator 3 at every frame start
- 0 The generator 3 will not reset in the starting point of a frame
- 1 The generator 3 will reset in the starting point of a frame

GF3[10:8] : The highest 3 bits of the generator 3 falling position (POR = 000)
GF3[7:0] : The lower byte of the generator 3 falling position (POR = 00000001)

GR3[10:8] : The highest 3 bits of the generator 3 rising position (POR = 000)
GR3[7:0] : The lower byte of the generator 3 rising position (POR = 00000000)

F[7] : Force the generator 3 output to 0 in non-display period
- 0 generator 3 is normal
- 1 generator 3 output is forced to 0 in non-display period

F[6:5] : Force the generator 3 output to 0 in odd or even lines
- 00 generator 3 is normal in both odd and even lines
- 01 generator 3 output is force to 0 in odd lines
- 10 generator 3 output is force to 0 in even lines
- 11 generator 3 is normal in both odd and even line

F[4:3] : Generator 3 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP3[10:8] : The highest 3 bits of the generator 3 period (POR = 100)
GP3[7:0] : The lower byte of the generator 3 period (POR = 00000000)

9.54 get_lcd_gen3

Command 0xC7
Parameters 7

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	0	1	1	1	C7
Parameter 1	1	A_7	0	0	0	0	0	0	0	xx
Parameter 2	1	0	0	0	0	0	$GF3_{10}$	$GF3_9$	$GF3_8$	xx
Parameter 3	1	$GF3_7$	$GF3_6$	$GF3_5$	$GF3_4$	$GF3_3$	$GF3_2$	$GF3_1$	$GF3_0$	xx
Parameter 4	1	0	0	0	0	0	$GR3_{10}$	$GR3_9$	$GR3_8$	xx
Parameter 5	1	$GR3_7$	$GR3_6$	$GR3_5$	$GR3_4$	$GR3_3$	$GR3_2$	$GR3_1$	$GR3_0$	xx
Parameter 6	1	F_7	F_6	F_5	F_4	F_3	$GP3_{10}$	$GP3_9$	$GP3_8$	xx
Parameter 7	1	$GP3_7$	$GP3_6$	$GP3_5$	$GP3_4$	$GP3_3$	$GP3_2$	$GP3_1$	$GP3_0$	xx

Description
Get the rise, fall, period and toggling properties of LCD signal generator 3

A[7] : Reset LCD generator 3 at every frame start
- 0 The generator 3 will not reset in the starting point of a frame
- 1 The generator 3 will reset in the starting point of a frame

GF3[10:8] : The highest 3 bits of the generator 3 falling position (POR = 000)
GF3[7:0] : The lower byte of the generator 3 falling position (POR = 00000001)

GR3[10:8] : The highest 3 bits of the generator 3 rising position (POR = 000)
GR3[7:0] : The lower byte of the generator 3 rising position (POR = 00000000)

F[7] : Force the generator 3 output to 0 in non-display period
- 0 generator 3 is normal
- 1 generator 3 output is forced to 0 in non-display period

F[6:5] : Force the generator 3 output to 0 in odd or even lines
- 00 generator 3 is normal in both odd and even lines
- 01 generator 3 output is force to 0 in odd lines
- 10 generator 3 output is force to 0 in even lines
- 11 generator 3 is normal in both odd and even line

F[4:3] : Generator 3 toggle mode
- 00 Disable
- 01 Toggle by pixel clock (LSHIFT)
- 10 Toggle by Line (LLINE)
- 11 Toggle by Frame (LFRAME)

GP3[10:8] : The highest 3 bits of the generator 3 period (POR = 100)
GP3[7:0] : The lower byte of the generator 3 period (POR = 00000000)

9.55 set_gpio0_rop

Command 0xC8
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	0	0	0	C8
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Set the GPIO0 with respect to the LCD signal generators using ROP operation. No effect if the GPIO0 is configured as general GPIO.

A[6:5] : Source 1 for GPIO0 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO0 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO0 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO0 (POR = 00000000)
Please refer to the Application note for the ROP operation

9.56 get_gpio0_rop

Command 0xC9
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	0	0	1	C9
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Get the GPIO0 properties with respect to the LCD signal generators.

A[6:5] : Source 1 for GPIO0 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO0 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO0 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO0 (POR = 00000000)
Please refer to the Application note for ROP operation

9.57 set_gpio1_rop

Command 0xCA
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	0	1	0	CA
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Set the GPIO1 with respect to the LCD signal generators using ROP operation. No effect if the GPIO1 is configured as general GPIO.

A[6:5] : Source 1 for GPIO1 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO1 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO1 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO1 (POR = 00000000)
Please refer to the Application note for the ROP operation

9.58 get_gpio1_rop

Command 0xCB
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	0	1	1	CB
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Get the GPIO1 properties with respect to the LCD signal generators.

A[6:5] : Source 1 for GPIO1 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO1 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO1 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO1 (POR = 00000000)
Please refer to the Application note for the ROP operation

9.59 set_gpio2_rop

Command 0xCC
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	1	0	0	CC
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Set the GPIO2 with respect to the LCD signal generators using ROP operation. No effect if the GPIO2 is configured as general GPIO.

A[6:5] : Source 1 for GPIO2 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO2 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO2 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO2 (POR = 00000000)
Please refer to the Application note for the ROP operation

9.60 get_gpio2_rop

Command 0xCD
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	1	0	1	CD
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Get the GPIO2 properties with respect to the LCD signal generators.

A[6:5] : Source 1 for GPIO2 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO2 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO2 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO2 (POR = 00000000)
Please refer to the Application note for the ROP operation

9.61 set_gpio3_rop

Command 0xCE
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	1	1	0	CE
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Set the GPIO3 with respect to the LCD signal generators using ROP operation. No effect if the GPIO3 is configured as general GPIO.

A[6:5] : Source 1 for GPIO3 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO3 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO3 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO3 (POR = 00000000)
Please refer to the Application note for the ROP operation

9.62 get_gpio3_rop

Command 0xCF
Parameters 2

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	0	1	1	1	1	CF
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	A_1	A_0	xx
Parameter 2	1	B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0	xx

Description
Get the GPIO3 properties with respect to the LCD signal generators.

A[6:5] : Source 1 for GPIO3 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[3:2] : Source 2 for GPIO3 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

A[1:0] : Source 3 for GPIO3 when controlled by LCDC (POR = 00)
- 00 Generator 0
- 01 Generator 1
- 10 Generator 2
- 11 Generator 3

B[7:0] : ROP operation to mux the source 1, 2 and 3 for GPIO3 (POR = 00000000)
Please refer to the Application note for the ROP operation

9.63 set_dbc_conf

Command 0xD0
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	1	0	0	0	0	D0
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	0	A_0	xx

Description
Set the Dynamic Backlight Control configuration.

A[6] : DBC Manual Brightness enable (POR = 1)
- 0 Enable
- 1 Disable

A[5] : Transition effect (POR = 0)
- 0 Transition effect disable
- 1 Transition effect enable

Transition effect is used to remove visible backlight flickering. If rapid brightness change is required, it is recommended to enable this bit.

A[3:2] : Energy saving selection for DBC (POR = 00)
- 00 DBC is disable
- 01 Conservative mode
- 10 Normal mode
- 11 Aggressive mode

A[0] : Master enable of DBC (POR = 0)
- 0 DBC disable
- 1 DBC enable

The hardware pin, PWM is the output signal from SSD1963 to the system backlight driver. So it should configure PWM module before enable DBC.

WRITE COMMAND "0xBE"
WRITE DATA "0x0E" (set PWM frequency)
WRITE DATA "0xFF" (dummy value if DBC is used)
WRITE DATA "0x09" (enable PWM controlled by DBC)
WRITE DATA "0xFF"
WRITE DATA "0x00"
WRITE DATA "0x00"

WRITE COMMAND "0xD4" (Define the threshold value)
WRITE DATA …..

WRITE COMMAND "0xD0"
WRITE DATA "0x0D" (Enable DBC with Aggressive mode)

9.64 get_dbc_conf

Command 0xD1
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	1	0	0	0	1	D1
Parameter 1	1	0	A_6	A_5	0	A_3	A_2	1	A_0	xx

Description
Get the current dynamic back light configuration.

A[6] : DBC Manual Brightness enable (POR = 1)
0 Enable
1 Disable

A[5] : Transition effect (POR = 0)
0 Transition effect disable
1 Transition effect enable

A[3:2] : Energy saving selection for DBC (POR = 00)
00 DBC is disable
01 Conservative mode
10 Normal mode
11 Aggressive mode

A[0] : Master enable DBC (POR = 0)
0 DBC disable
1 DBC enable

9.65 set_dbc_th

Command 0xD4
Parameters 9

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	1	0	1	0	0	D4
Parameter 1	1	0	0	0	0	0	0	0	DBC_TH1_{16}	xx
Parameter 2	1	DBC_TH1_{15}	DBC_TH1_{14}	DBC_TH1_{13}	DBC_TH1_{12}	DBC_TH1_{11}	DBC_TH1_{10}	DBC_TH1_{9}	DBC_TH1_{8}	xx
Parameter 3	1	DBC_TH1_{7}	DBC_TH1_{6}	DBC_TH1_{5}	DBC_TH1_{4}	DBC_TH1_{3}	DBC_TH1_{2}	DBC_TH1_{1}	DBC_TH1_{0}	xx
Parameter 4	1	0	0	0	0	0	0	0	DBC_TH2_{16}	xx
Parameter 5	1	DBC_TH2_{15}	DBC_TH2_{14}	DBC_TH2_{13}	DBC_TH2_{12}	DBC_TH2_{11}	DBC_TH2_{10}	DBC_TH2_{9}	DBC_TH2_{8}	xx
Parameter 6	1	DBC_TH2_{7}	DBC_TH2_{6}	DBC_TH2_{5}	DBC_TH2_{4}	DBC_TH2_{3}	DBC_TH2_{2}	DBC_TH2_{1}	DBC_TH2_{0}	xx
Parameter 7	1	0	0	0	0	0	0	0	DBC_TH3_{16}	xx
Parameter 8	1	DBC_TH3_{15}	DBC_TH3_{14}	DBC_TH3_{13}	DBC_TH3_{12}	DBC_TH3_{11}	DBC_TH3_{10}	DBC_TH3_{9}	DBC_TH3_{8}	xx
Parameter 9	1	DBC_TH3_{7}	DBC_TH3_{6}	DBC_TH3_{5}	DBC_TH3_{4}	DBC_TH3_{3}	DBC_TH3_{2}	DBC_TH3_{1}	DBC_TH3_{0}	xx

Description
Set the threshold for each level of power saving.

DBC_TH1[16] : High byte of the threshold setting for the Conservative mode of DBC. (POR = 0)
DBC_TH1[15:8] : 2nd byte of the threshold setting for the Conservative mode of DBC. (POR = 00000000)
DBC_TH1[7:0] : Low byte of the threshold setting for the Conservative mode of DBC. (POR = 00000000)

TH1 = display width * display height * 3 * 0.1 /16

DBC_TH2[16] : High byte of the threshold setting for the Normal mode of DBC. (POR = 0)
DBC_TH2[15:8] : 2nd byte of the threshold setting for the Normal mode of DBC. (POR = 00000000)
DBC_TH2[7:0] : Low byte of the threshold setting for the Normal mode of DBC. (POR = 00000000)

TH2 = display width * display height * 3 * 0.25 /16

DBC_TH3[16] : High byte of the threshold setting for the Aggressive mode of DBC. (POR = 0)
DBC_TH3[15:8] : 2nd byte of the threshold setting for the Aggressive mode of DBC. (POR = 00000000)
DBC_TH3[7:0] : Low byte of the threshold setting for the Aggressive mode of DBC. (POR = 00000000)

TH3 = display width * display height * 3 * 0.6 /16

9.66 get_dbc_th

Command 0xD5
Parameters 9

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	0	1	0	1	0	1	D5
Parameter 1	1	0	0	0	0	0	0	0	DBC_TH1_{16}	xx
Parameter 2	1	DBC_TH1_{15}	DBC_TH1_{14}	DBC_TH1_{13}	DBC_TH1_{12}	DBC_TH1_{11}	DBC_TH1_{10}	DBC_TH1_{9}	DBC_TH1_{8}	xx
Parameter 3	1	DBC_TH1_{7}	DBC_TH1_{6}	DBC_TH1_{5}	DBC_TH1_{4}	DBC_TH1_{3}	DBC_TH1_{2}	DBC_TH1_{1}	DBC_TH1_{0}	xx
Parameter 4	1	0	0	0	0	0	0	0	DBC_TH2_{16}	xx
Parameter 5	1	DBC_TH2_{15}	DBC_TH2_{14}	DBC_TH2_{13}	DBC_TH2_{12}	DBC_TH2_{11}	DBC_TH2_{10}	DBC_TH2_{9}	DBC_TH2_{8}	xx
Parameter 6	1	DBC_TH2_{7}	DBC_TH2_{6}	DBC_TH2_{5}	DBC_TH2_{4}	DBC_TH2_{3}	DBC_TH2_{2}	DBC_TH2_{1}	DBC_TH2_{0}	xx
Parameter 7	1	0	0	0	0	0	0	0	DBC_TH3_{16}	xx
Parameter 8	1	DBC_TH3_{15}	DBC_TH3_{14}	DBC_TH3_{13}	DBC_TH3_{12}	DBC_TH3_{11}	DBC_TH3_{10}	DBC_TH3_{9}	DBC_TH3_{8}	xx
Parameter 9	1	DBC_TH3_{7}	DBC_TH3_{6}	DBC_TH3_{5}	DBC_TH3_{4}	DBC_TH3_{3}	DBC_TH3_{2}	DBC_TH3_{1}	DBC_TH3_{0}	xx

Description
Get the threshold for each level of power saving.

DBC_TH1[16] : High byte of the threshold setting for the Conservative mode of DBC. (POR = 0)
DBC_TH1[15:8] : 2nd byte of the threshold setting for the Conservative mode of DBC. (POR = 00000000)
DBC_TH1[7:0] : Low byte of the threshold setting for the Conservative mode of DBC. (POR = 00000000)

DBC_TH2[16] : High byte of the threshold setting for the Normal mode of DBC. (POR = 0)
DBC_TH2[15:8] : 2nd byte of the threshold setting for the Normal mode of DBC. (POR = 00000000)
DBC_TH2[7:0] : Low byte of the threshold setting for the Normal mode of DBC. (POR = 00000000)

DBC_TH3[16] : High byte of the threshold setting for the Aggressive mode of DBC. (POR = 0)
DBC_TH3[15:8] : 2nd byte of the threshold setting for the Aggressive mode of DBC. (POR = 00000000)
DBC_TH3[7:0] : Low byte of the threshold setting for the Aggressive mode of DBC. (POR = 00000000)

9.67 set_pll

Command 0xE0
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	0	0	0	0	0	E0
Parameter 1	1	0	0	0	0	0	0	A_1	A_0	xx

Description
Start the PLL. Before the start, the system was operated with the crystal oscillator or clock input.

A[1] : Lock PLL (POR = 0)
After PLL enabled for 100us, can start to lock PLL
0 Use reference clock as system clock
1 Use PLL output as system clock

A[0] : Enable PLL (POR = 0)
0 Disable PLL
1 Enable PLL

Before enabling PLL, the PLL setting (“0xE2”) have to be configured first. After PLL enabled for 100us, can start to lock PLL. SSD1963 needed to switch to PLL output as system clock after PLL is locked. The following is the program sequence.

WRITE COMMAND “0xE0”
WRITE DATA “0x01”
Wait 100us to let the PLL stable
WRITE COMMAND “0xE0”
WRITE DATA “0x03”
WRITE COMMAND “0x01”

* Note : SSD1963 is operating under reference clock before PLL is locked, registers cannot be set faster than half of the reference clock frequency. For instance, SSD1963 with a 10MHz reference clock is not allowed to be programmed higher than 5M words/s.

9.68 set_pll_mn

Command 0xE2
Parameters 3

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	0	0	0	1	0	E2
Parameter 1	1	M_7	M_6	M_5	M_4	M_3	M_2	M_1	M_0	xx
Parameter 2	1	0	0	0	0	N_3	N_2	N_1	N_0	xx
Parameter 3	1	0	0	0	0	0	C_2	0	0	xx

Description
Set the MN of PLL
M[7:0] : Multiplier (M) of PLL. (POR = 00101101)

N[3:0] : Divider (N) of PLL. (POR = 0011)

C[2] : Effectuate MN value (POR = 0)
0 Ignore the multiplier (N) and divider (N) values
1 Effectuate the multiplier and divider value

VCO = Reference input clock x (M + 1)
PLL frequency = VCO / (N + 1)

*** Note : 250MHz < VCO < 800MHz**

For a 10MHz reference clock to obtain 100MHz PLL frequency, user cannot program M = 19 and N = 1. The setting in this situation is setting M=29 and N=2, where 10 x 30 / 3 = 100MHz.

WRITE COMMAND “0xE2”
WRITE DATA “0x1D” (M=29)
WRITE DATA “0x02” (N=2)
WRITE DATA “0x54” (Dummy Byte)

9.69 get_pll_mn

Command 0xE3
Parameters 3

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	0	0	0	1	1	E3
Parameter 1	1	M_7	M_6	M_5	M_4	M_3	M_2	M_1	M_0	xx
Parameter 2	1	0	0	0	0	N_3	N_2	N_1	N_0	xx
Parameter 3	1	0	0	0	0	0	C_2	0	0	xx

Description
Get the MN setting of PLL

M[7:0] : Multiplier (M) of PLL. (POR = 00101101)

N[3:0] : Divider (N) of PLL. (POR = 0011)

C[2] : Effectuate MN value (POR = 0)
0 Ignore the multiplier (M) and divider (N) values.
1 Effectuate the multiplier and divider value

9.70 get_pll_status

Command 0xE4
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	0	0	1	0	0	E4
Parameter 1	1	0	0	0	0	0	A_2	0	0	xx

Description
Get the PLL status
A[2] : PLL Lock
0 Not locked
1 Locked

9.71 set_deep_sleep

Command 0xE5
Parameters None

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	0	0	1	0	1	E5

Description
Set deep sleep mode. PLL would be stopped.

It needs to issue 2 dummy read to exit Deep Sleep mode.

9.72 set_lshift_freq

Command 0xE6
Parameters 3

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	0	0	1	1	0	E6
Parameter 1	1	0	0	0	0	$LCDC_FPR_{19}$	$LCDC_FPR_{18}$	$LCDC_FPR_{17}$	$LCDC_FPR_{16}$	xx
Parameter 2	1	$LCDC_FPR_{15}$	$LCDC_FPR_{14}$	$LCDC_FPR_{13}$	$LCDC_FPR_{12}$	$LCDC_FPR_{11}$	$LCDC_FPR_{10}$	$LCDC_FPR_{9}$	$LCDC_FPR_{8}$	xx
Parameter 3	1	$LCDC_FPR_{7}$	$LCDC_FPR_{6}$	$LCDC_FPR_{5}$	$LCDC_FPR_{4}$	$LCDC_FPR_{3}$	$LCDC_FPR_{2}$	$LCDC_FPR_{1}$	$LCDC_FPR_{0}$	xx

Description
Set the LSHIFT (pixel clock) frequency

LCDC_FPR[19:16] : The highest 4 bits for the pixel clock frequency settings. (POR = 0111)
LCDC_FPR[15:8] : The higher byte for the pixel clock frequency settings. (POR = 11111111)
LCDC_FPR[7:0] : The low byte for the pixel clock frequency settings. (POR = 11111111)

For parallel LCD interface:
Configure the pixel clock to PLL freq x ((LCDC_FPR + 1) / 2^{20})

To obtain PCLK = 5.3MHz with PLL Frequency = 100MHz,
5.3MHz = 100MHz * (LCDC_FPR+ 1) / 2^{20}
LCDC_FPR = 55574

WRITE COMMAND “0xE6”
WRITE DATA “0x00” (LCDC_FPR = 55574)
WRITE DATA “0xD9”
WRITE DATA “0x16”

For serial LCD interface:
Configure the pixel clock to PLL freq x ((LCDC_FPR + 1) / 2^{20}) *4

To obtain PCLK = 5.3MHz with PLL Frequency = 100MHz,
5.3MHz = 100MHz * ((LCDC_FPR+ 1) / 2^{20})*4
LCDC_FPR = 13892

WRITE COMMAND “0xE6”
WRITE DATA “0x00” (LCDC_FPR = 13892)
WRITE DATA “0x36”
WRITE DATA “0x44”

9.73 get_lshift_freq

Command 0xE7
Parameters 3

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	0	0	1	1	1	E7
Parameter 1	1	0	0	0	0	$LCDC_FPR_{19}$	$LCDC_FPR_{18}$	$LCDC_FPR_{17}$	$LCDC_FPR_{16}$	xx
Parameter 2	1	$LCDC_FPR_{15}$	$LCDC_FPR_{14}$	$LCDC_FPR_{13}$	$LCDC_FPR_{12}$	$LCDC_FPR_{11}$	$LCDC_FPR_{10}$	$LCDC_FPR_{9}$	$LCDC_FPR_{8}$	xx
Parameter 3	1	$LCDC_FPR_{7}$	$LCDC_FPR_{6}$	$LCDC_FPR_{5}$	$LCDC_FPR_{4}$	$LCDC_FPR_{3}$	$LCDC_FPR_{2}$	$LCDC_FPR_{1}$	$LCDC_FPR_{0}$	xx

Description
Get the current LSHIFT (pixel clock) frequency setting

LCDC_FPR[19:16] : The highest 4 bits for the pixel clock frequency settings. (POR = 0111)
LCDC_FPR[15:8] : The higher byte for the pixel clock frequency settings. (POR = 11111111)
LCDC_FPR[7:0] : The low byte for the pixel clock frequency settings. (POR = 11111111)

9.74 set_pixel_data_interface

Command 0xF0
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	1	0	0	0	0	F0
Parameter 1	1	0	0	0	0	0	A_2	A_1	A_0	xx

Description
Set the pixel data format to 8-bit / 9-bit / 12-bit / 16-bit / 16-bit(565) / 18-bit / 24-bit in the parallel host processor interface. This command is used for display data only, the command format is always 8 bit.

A[2:0] : Pixel Data Interface Format (POR = 101)

000	8-bit
001	12-bit
010	16-bit packed
011	16-bit (565 format)
100	18-bit
101	24-bit
110	9-bit
Others	Reserved

* Note : The un-used data bus will be driven to ground by SSD1963, so don't connect the un-used data bus to MCU.

9.75 get_pixel_data_interface

Command 0xF1
Parameters 1

	D/C	D7	D6	D5	D4	D3	D2	D1	D0	Hex
Command	0	1	1	1	1	0	0	0	1	F1
Parameter 1	1	0	0	0	0	0	A_2	A_1	A_0	xx

Description
Get the current pixel data format settings in the parallel host processor interface.

A[2:0] : Pixel Data Interface Format (POR = 101)

000	8-bit
001	12-bit
010	16-bit packed
011	16-bit (565 format)
100	18-bit
101	24-bit
110	9-bit
Others	Reserved

10 MAXIMUM RATINGS

Table 10-1: Maximum Ratings (Voltage Referenced to V_{SS})

Symbol	Parameter	Value	Unit
V_{DDD}	Digital Core power supply	-0.5 to 1.8	V
V_{DDPLL}	PLL power supply	-0.5 to 1.8	V
V_{DDLCD}	LCD Interface power supply	-0.5 to 4.6	V
V_{DDIO}	I/O power supply	-0.5 to 4.6	V
V_{IN}	Input Voltage	-0.5 to 4.6	V
V_{OUT}	Output Voltage	-0.5 to 4.6	V
T_{SOL}	Solder Temperature / Time	225 for 40 sec max at solder ball	°C
T_{STG}	Storage temperature	-45 to 125	°C
T_A	Operating temperature	-30 to 85	°C

Maximum ratings are those values beyond which damages to the device may occur. Functional operation should be restricted to the limits in the Electrical Characteristics tables or Pin Description section

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high impedance circuit. For proper operation it is recommended that V_{IN} and V_{OUT} be constrained to the range $V_{SS} < (V_{IN}$ or $V_{OUT}) < V_{DDIO}$. Reliability of operation is enhanced if unused input is connected to an appropriate logic voltage level (e.g., either V_{SS} or V_{DDIO}). Unused outputs must be left open. This device may be light sensitive. Caution should be taken to avoid exposure of this device to any light source during normal operation. This device is not radiation protected.

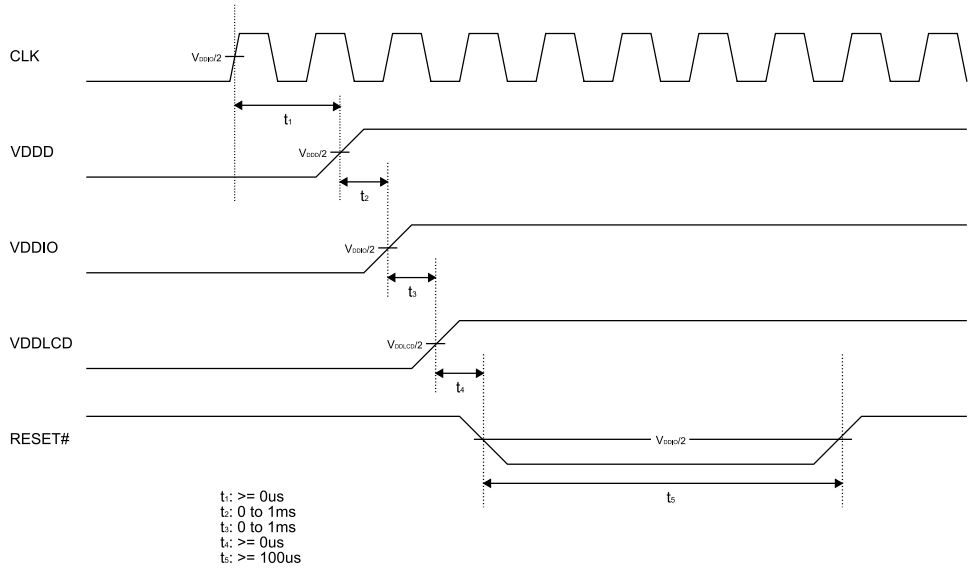
11 RECOMMENDED OPERATING CONDITIONS

Table 11-1: Recommended Operating Condition

Symbol	Parameter	Min	Typ	Max	Unit
V_{DDD}	Digital Core power supply	1.10	1.2	1.30	V
V_{DDPLL}	PLL power supply	1.10	1.2	1.30	V
V_{DDLCD}	LCD Interface power supply	1.65	3.3	3.6	V
V_{DDIO}	I/O power supply	1.65	3.3	3.6	V

11.1 Power-up sequence

Figure 11-1: Power-up Sequence



Note

Clock reference is only applicable when CLK is used.

12 DC CHARACTERISTICS

Conditions:

Voltage referenced to V_{SS}
V_{DDD}, V_{DDPLL} = 1.2V
V_{DDIO}, V_{DDLCD} = 3.3V
T_A = 25°C

Table 12-1 : DC Characteristics

Symbol	Parameter	Test Condition	Min	Typ	Max	Unit
PSTY	Quiescent Power			300	500	uW
IIZ	Input leakage current		-1		1	uA
IOZ	Output leakage current		-1		1	uA
VOH	Output high voltage		$0.8V_{DDIO}$			V
VOL	Output low voltage				$0.2V_{DDIO}$	V
VIH	Input high voltage		$0.8V_{DDIO}$		V_{DDIO} + 0.5	V
VIL	Input low voltage				$0.2V_{DDIO}$	V

13 AC CHARACTERISTICS

Conditions:

Voltage referenced to V_{SS}
V_{DDD}, V_{DDPLL} = 1.2V
V_{DDIO}, V_{DDLCD} = 3.3V
T_A = 25°C
C_L = 50pF (Bus/CPU Interface)
C_L = 0pF (LCD Panel Interface)

13.1 Clock Timing

Table 13-1: Clock Input Requirements for CLK (PLL-bypass)

Symbol	Parameter	Min	Max	Units
F_{CLK}	Input Clock Frequency (CLK)		110	MHz
T_{CLK}	Input Clock period (CLK)	$1/f_{CLK}$		ns

Table 13-2 : Clock Input Requirements for CLK

Symbol	Parameter	Min	Max	Units
F_{CLK}	Input Clock Frequency (CLK)	2.5	50	MHz
T_{CLK}	Input Clock period (CLK)	$1/f_{CLK}$		ns

Table 13-3 : Clock Input Requirements for crystal oscillator XTAL

Symbol	Parameter	Min	Max	Units
F_{XTAL}	Input Clock Frequency	2.5	10	MHz
T_{XTAL}	Input Clock period	$1/f_{XTAL}$		ns

13.2 MCU Interface Timing

13.2.1 Parallel 6800-series Interface Timing

Table 13-4: Parallel 6800-series Interface Timing Characteristics (Use CS# as clock)

Symbol	Parameter		Min	Typ	Max	Unit
f_{MCLK}	System Clock Frequency*		1	-	110	MHz
t_{MCLK}	System Clock Period*		$1/f_{MCLK}$	-	-	ns
t_{PWCSH}	Control Pulse High Width	Write	13	1.5* t_{MCLK}	-	ns
		Read	30	3.5* t_{MCLK}		
t_{PWCSL}	Control Pulse Low Width	Write (next write cycle)	13	1.5* t_{MCLK}	-	ns
		Write (next read cycle)	80	9* t_{MCLK}		
		Read	80	9* t_{MCLK}		
t_{AS}	Address Setup Time		2	-	-	ns
t_{AH}	Address Hold Time		2	-	-	ns
t_{DSW}	Data Setup Time		4	-	-	ns
t_{DHW}	Data Hold Time		1	-	-	ns
t_{PLW}	Write Low Time		14	-	-	ns
t_{PHW}	Write High Time		14	-	-	ns
t_{PLWR}	Read Low Time		38	-	-	ns
t_{ACC}	Data Access Time		32	-	-	ns
t_{DHR}	Output Hold time		1	-	-	ns
t_R	Rise Time		-	-	0.5	ns
t_F	Fall Time		-	-	0.5	ns

* System Clock denotes external input clock (PLL-bypass) or internal generated clock (PLL-enabled)

Figure 13-1: Parallel 6800-series Interface Timing Diagram (Use CS# as Clock)

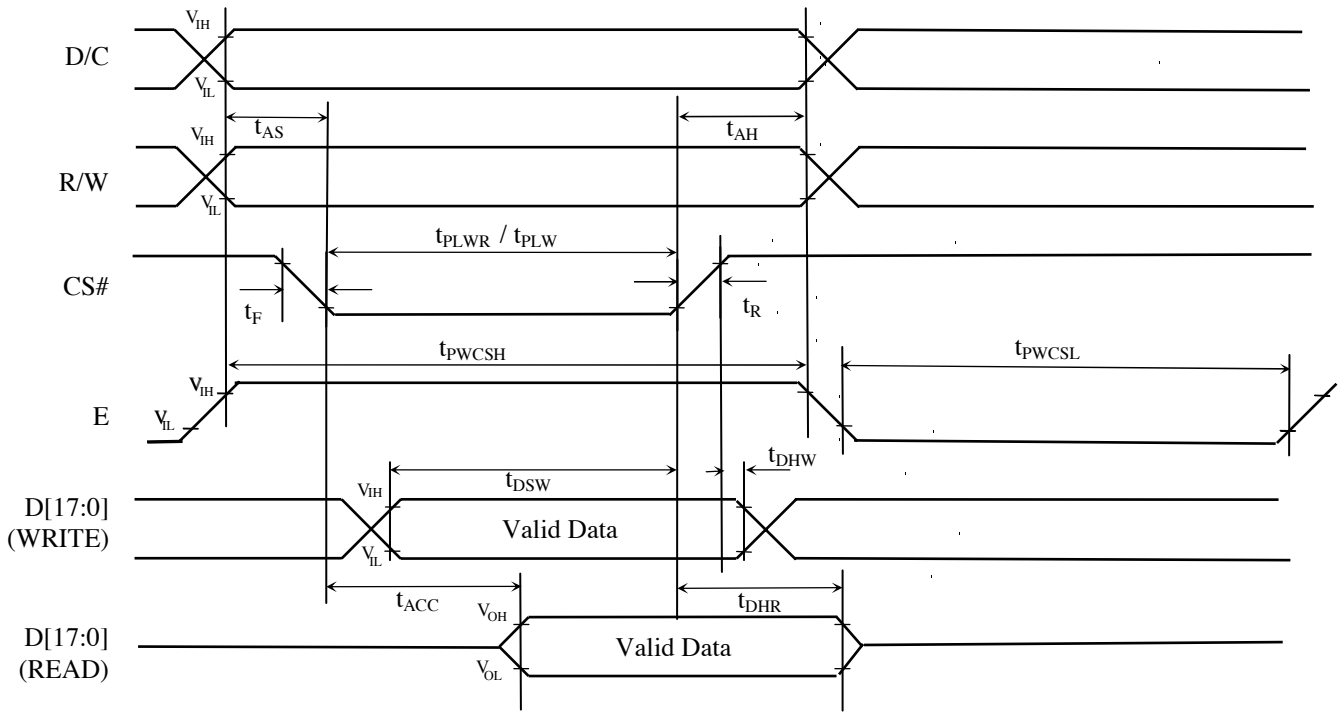
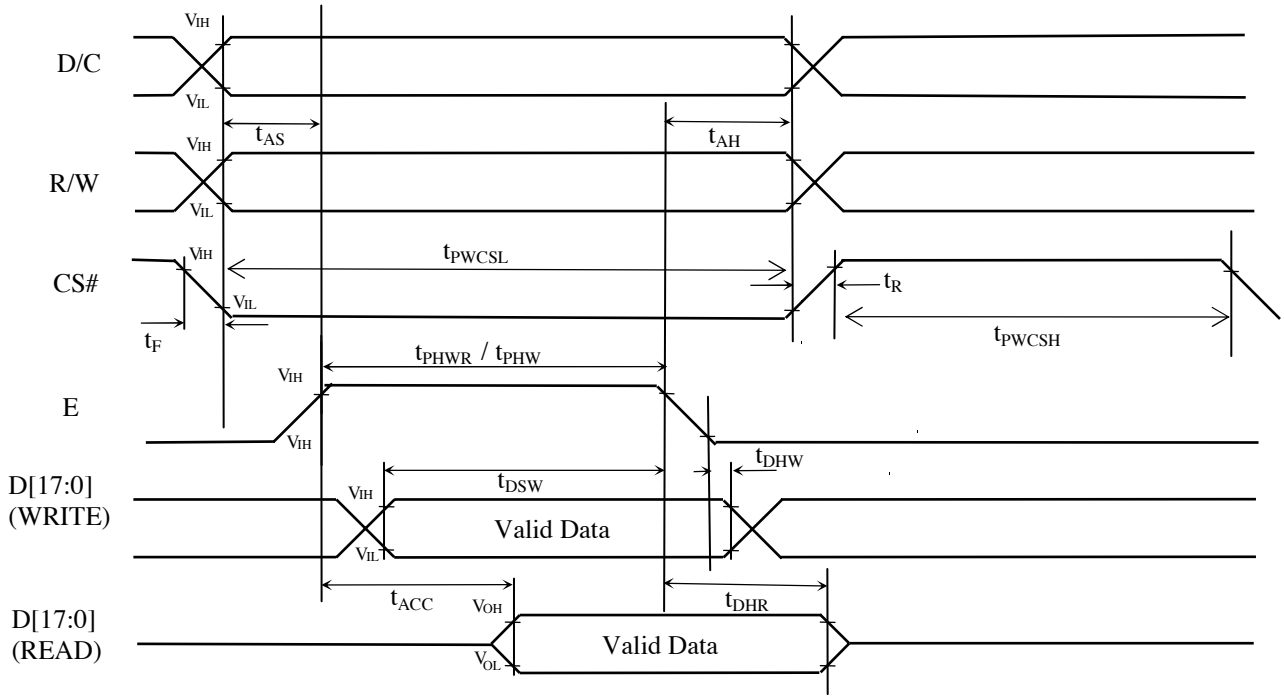


Table 13-5: Parallel 6800-series Interface Timing Characteristics (Use E as clock)

Symbol	Parameter		Min	Typ	Max	Unit
f_{MCLK}	System Clock Frequency*		1	-	110	MHz
t_{MCLK}	System Clock Period*		1/ f_{MCLK}	-	-	ns
t_{PWCSH}	Control Pulse Low Width	Write (next write cycle) Write (next read cycle) Read	13 80 80	1.5* t_{MCLK} 9* t_{MCLK} 9* t_{MCLK}	-	ns
t_{PWCSL}	Control Pulse High Width	Write Read	13 30	1.5* t_{MCLK} 3.5* t_{MCLK}	-	ns
t_{AS}	Address Setup Time		2	-	-	ns
t_{AH}	Address Hold Time		2	-	-	ns
t_{DSW}	Data Setup Time		4	-	-	ns
t_{DHW}	Data Hold Time		1	-	-	ns
t_{PLW}	Write Low Time		14	-	-	ns
t_{PHW}	Write High Time		14	-	-	ns
t_{PLWR}	Read Low Time		38	-	-	ns
t_{ACC}	Data Access Time		32	-	-	ns
t_{DHR}	Output Hold time		1	-	-	ns
t_R	Rise Time		-	-	0.5	ns
t_F	Fall Time		-	-	0.5	ns

* System Clock denotes external input clock (PLL-bypass) or internal generated clock (PLL-enabled)

Figure 13-2: Parallel 6800-series Interface Timing Diagram (Use E as Clock)



13.2.2 Parallel 8080-series Interface Timing

Table 13-6: Parallel 8080-series Interface Timing Characteristics

Symbol	Parameter		Min	Typ	Max	Unit
f_{MCLK}	System Clock Frequency*		1	-	110	MHz
t_{MCLK}	System Clock Period*		$1/ f_{MCLK}$	-	-	ns
t_{PWCSL}	Control Pulse High Width	Write Read	13 30	$1.5* t_{MCLK}$ $3.5* t_{MCLK}$	-	ns
t_{PWCSH}	Control Pulse Low Width	Write (next write cycle) Write (next read cycle) Read	13 80 80	$1.5* t_{MCLK}$ $9* t_{MCLK}$ $9* t_{MCLK}$	-	ns
t_{AS}	Address Setup Time		1	-	-	ns
t_{AH}	Address Hold Time		2	-	-	ns
t_{DSW}	Write Data Setup Time		4	-	-	ns
t_{DHW}	Write Data Hold Time		1	-	-	ns
t_{PWLW}	Write Low Time		12	-	-	ns
t_{DHR}	Read Data Hold Time		1	-	-	ns
t_{ACC}	Access Time		32	-	-	ns
t_{PWLR}	Read Low Time		36	-	-	ns
t_R	Rise Time		-	-	0.5	ns
t_F	Fall Time		-	-	0.5	ns
t_{CS}	Chip select setup time		2	-	-	ns
t_{CSH}	Chip select hold time to read signal		3	-	-	ns

* System Clock denotes external input clock (PLL-bypass) or internal generated clock (PLL-enabled)

Figure 13-3: Parallel 8080-series Interface Timing Diagram (Write Cycle)

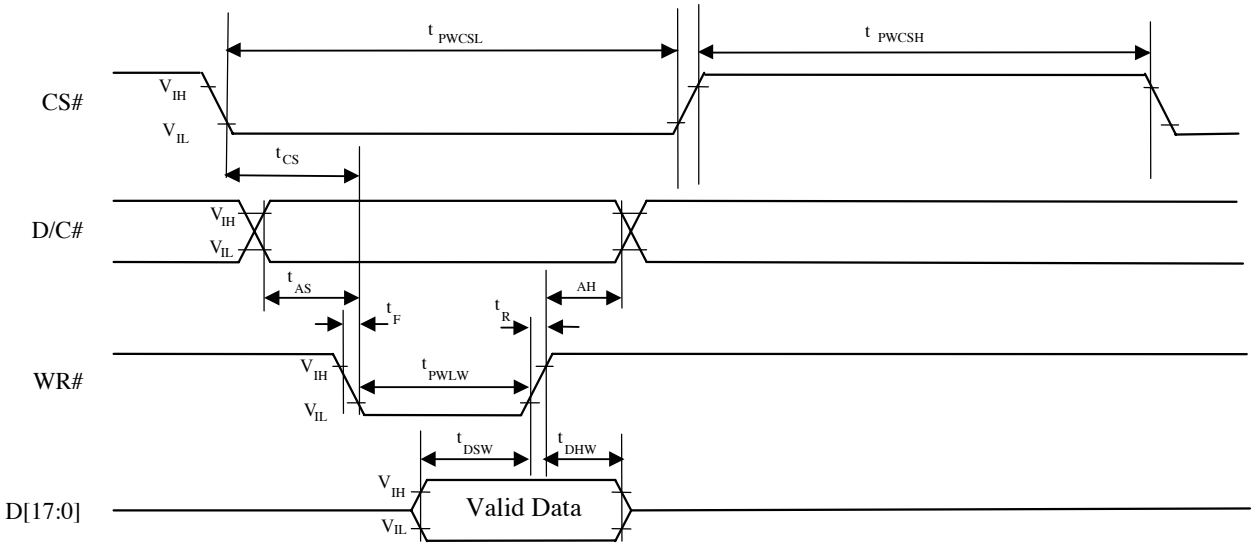
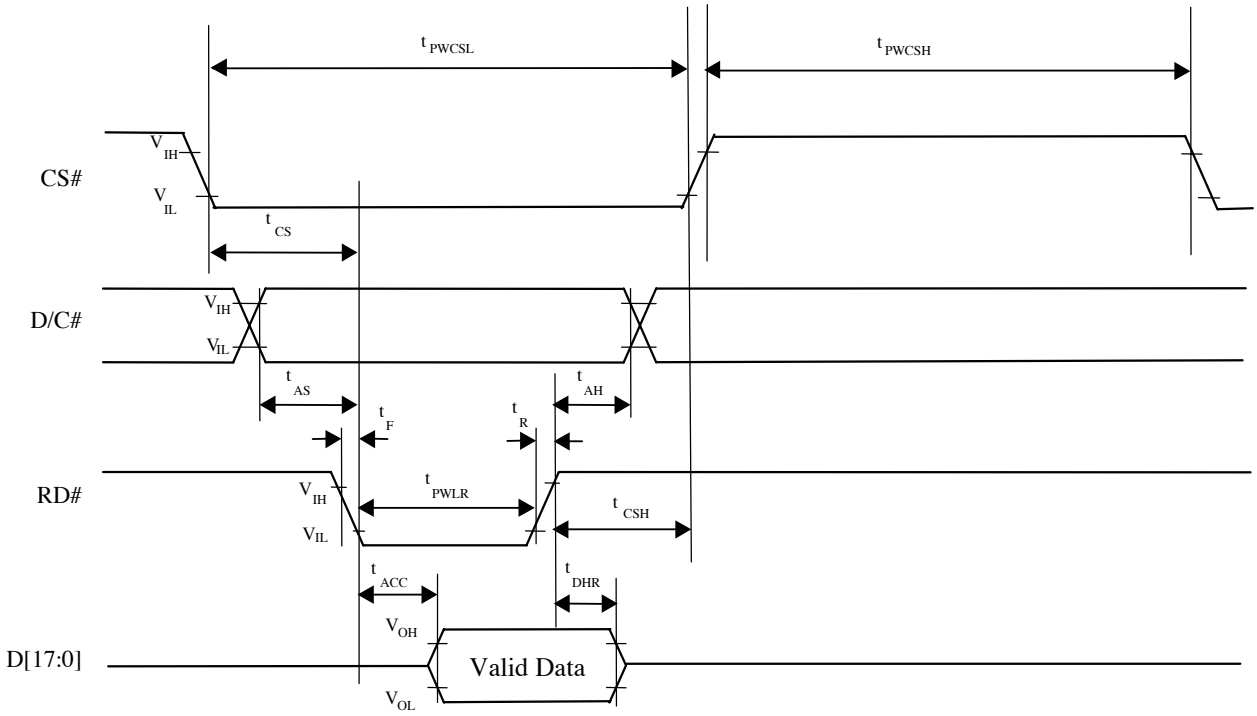
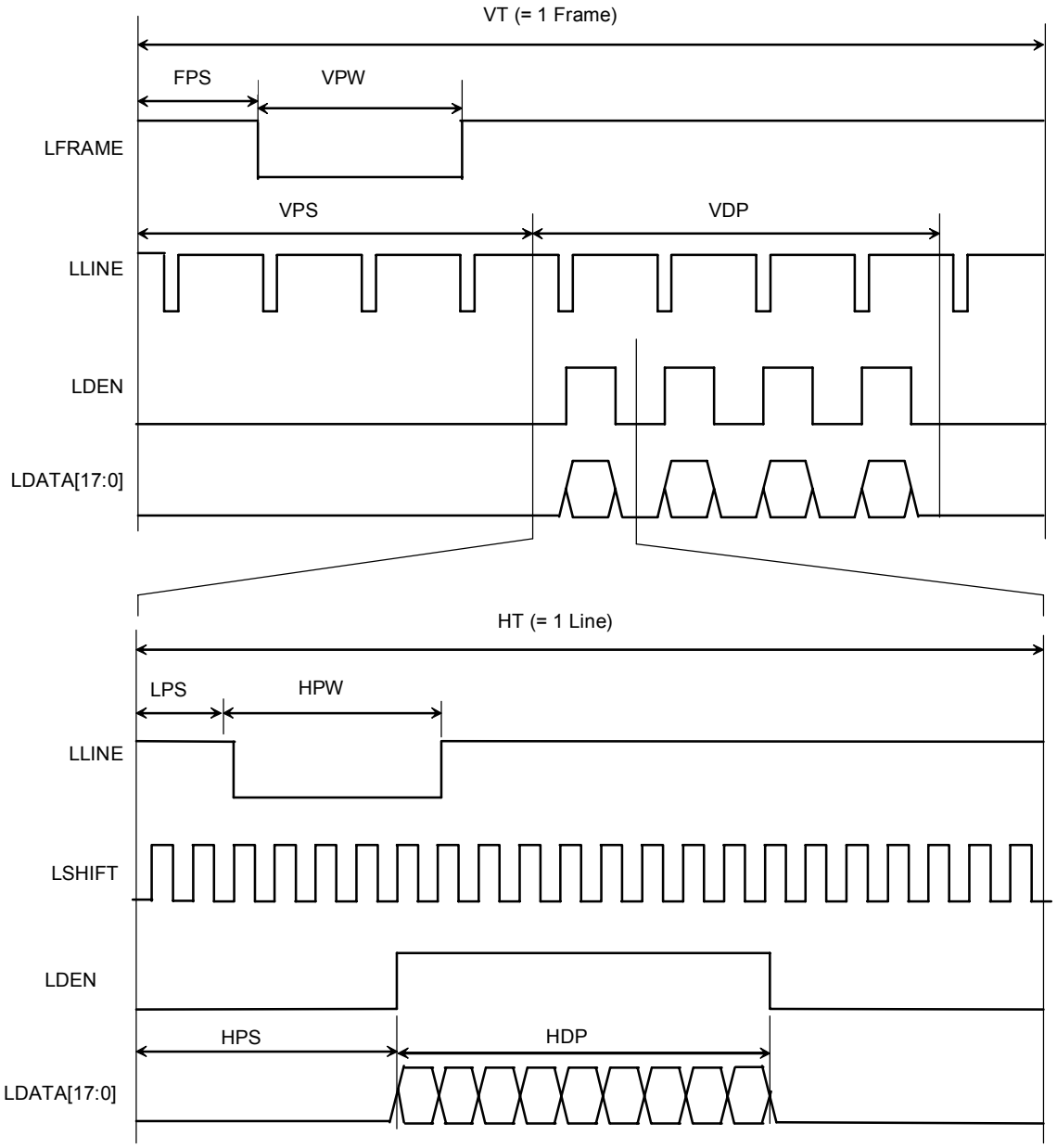


Figure 13-4: Parallel 8080-series Interface Timing Diagram (Read Cycle)



13.3 Parallel LCD Interface Timing

Figure 13-5: Generic TFT Panel Timing



13.4 Serial RGB Interface Timing

Figure 13-6: Serial RGB Interface Timing (without dummy mode)

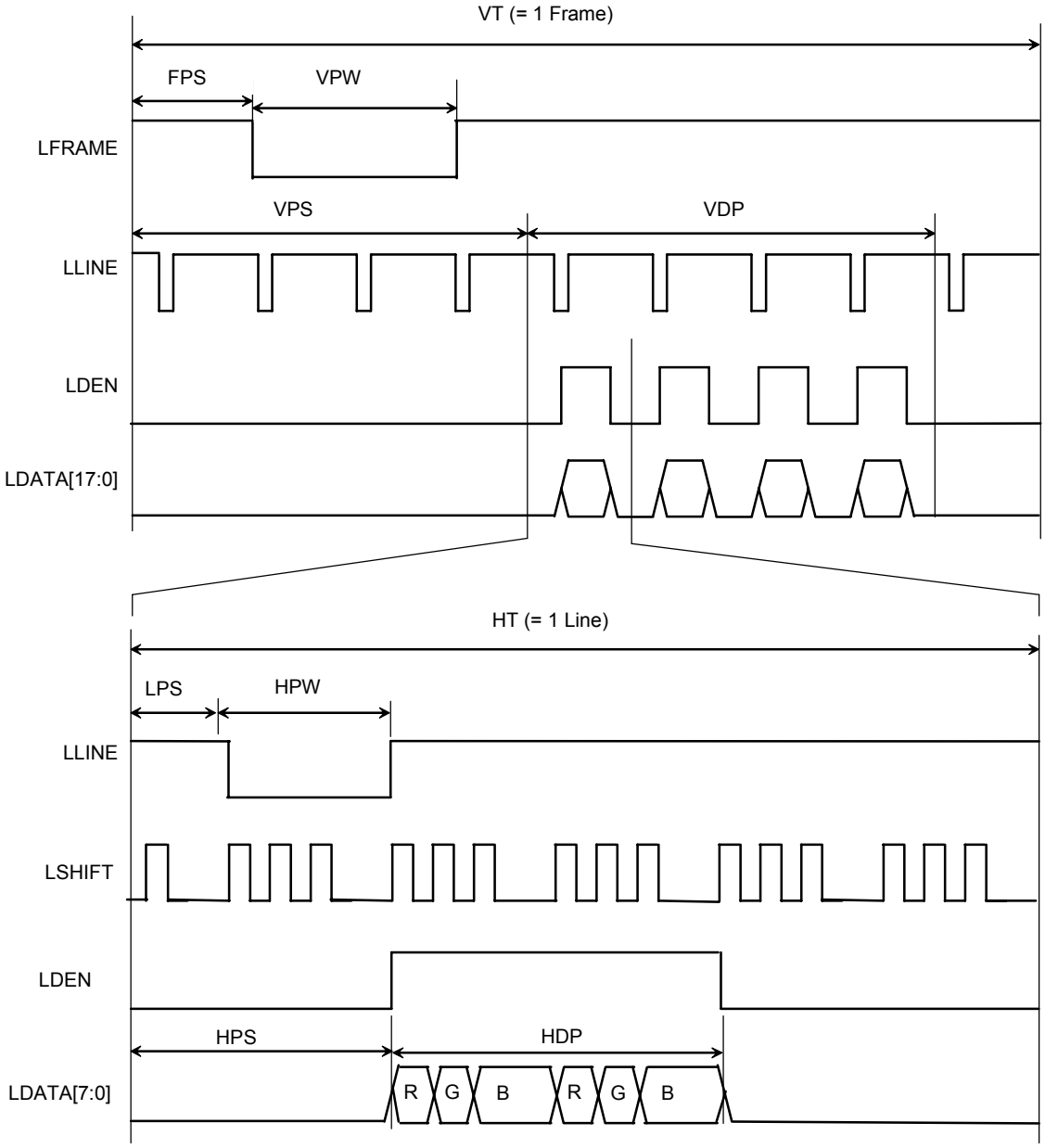
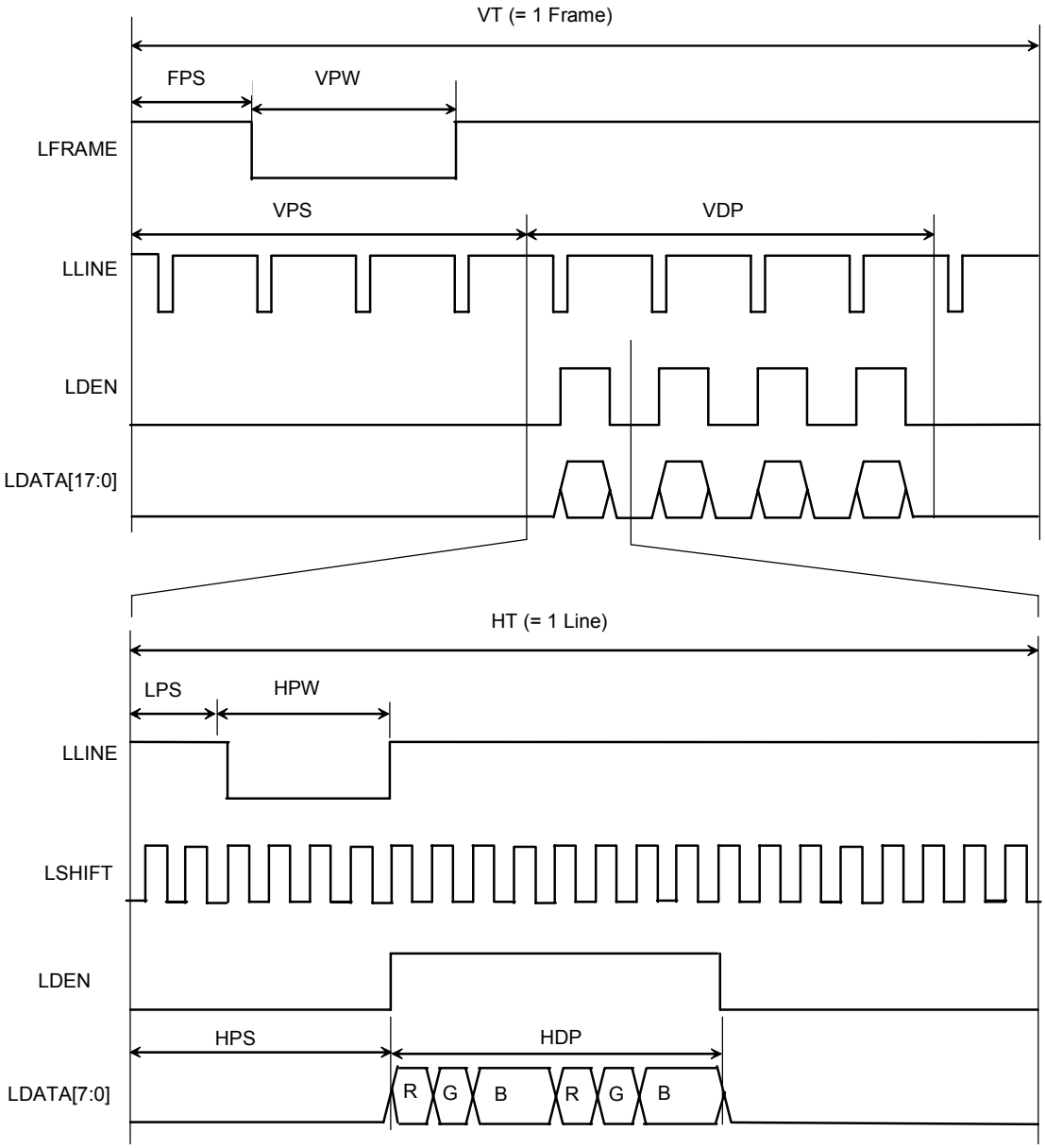


Figure 13-7: Serial RGB Interface Timing (with dummy mode)



14 APPLICATION EXAMPLE

Figure 14-1 : Application circuit for SSD1963 (With Direct clock input)

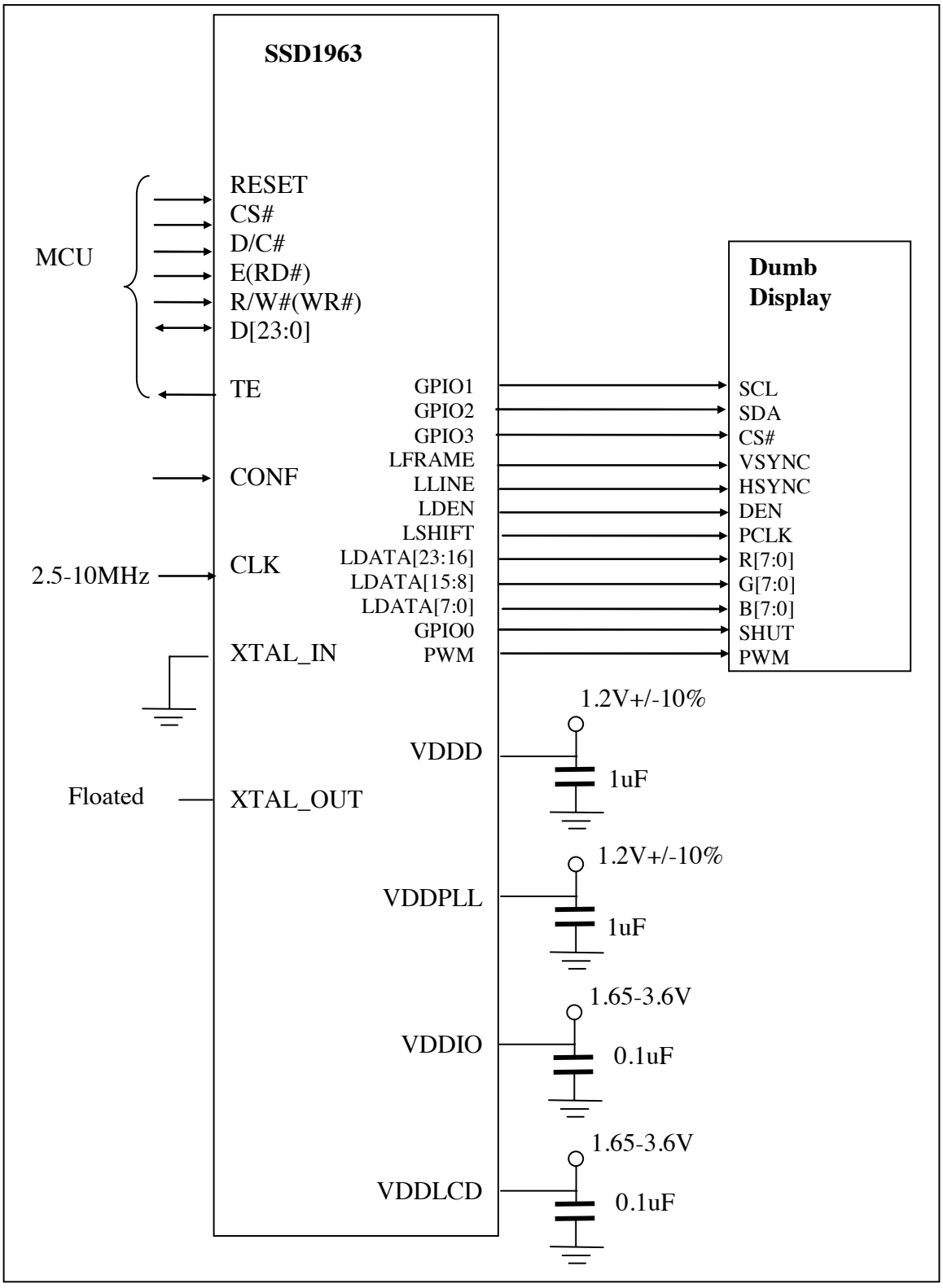
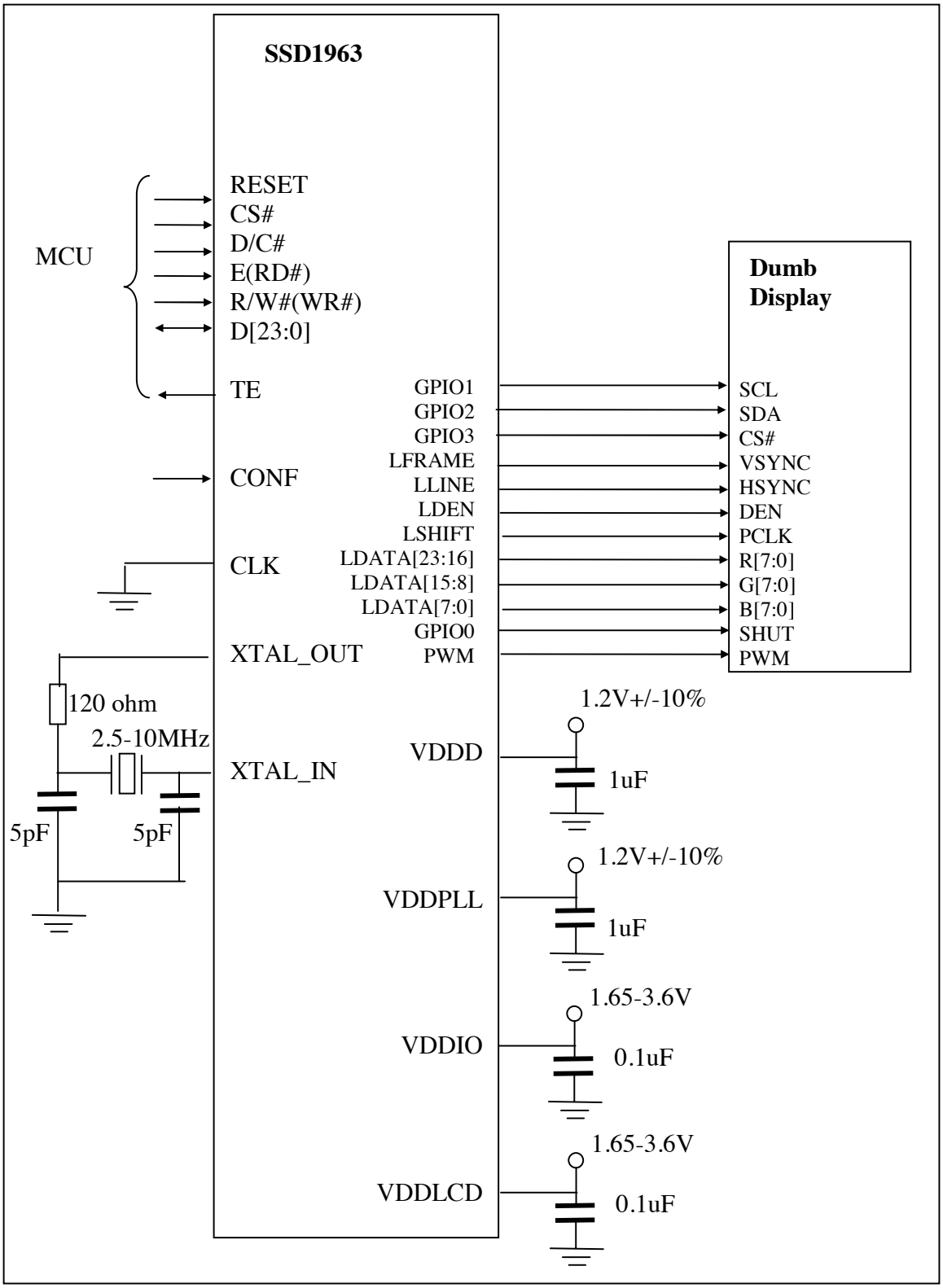
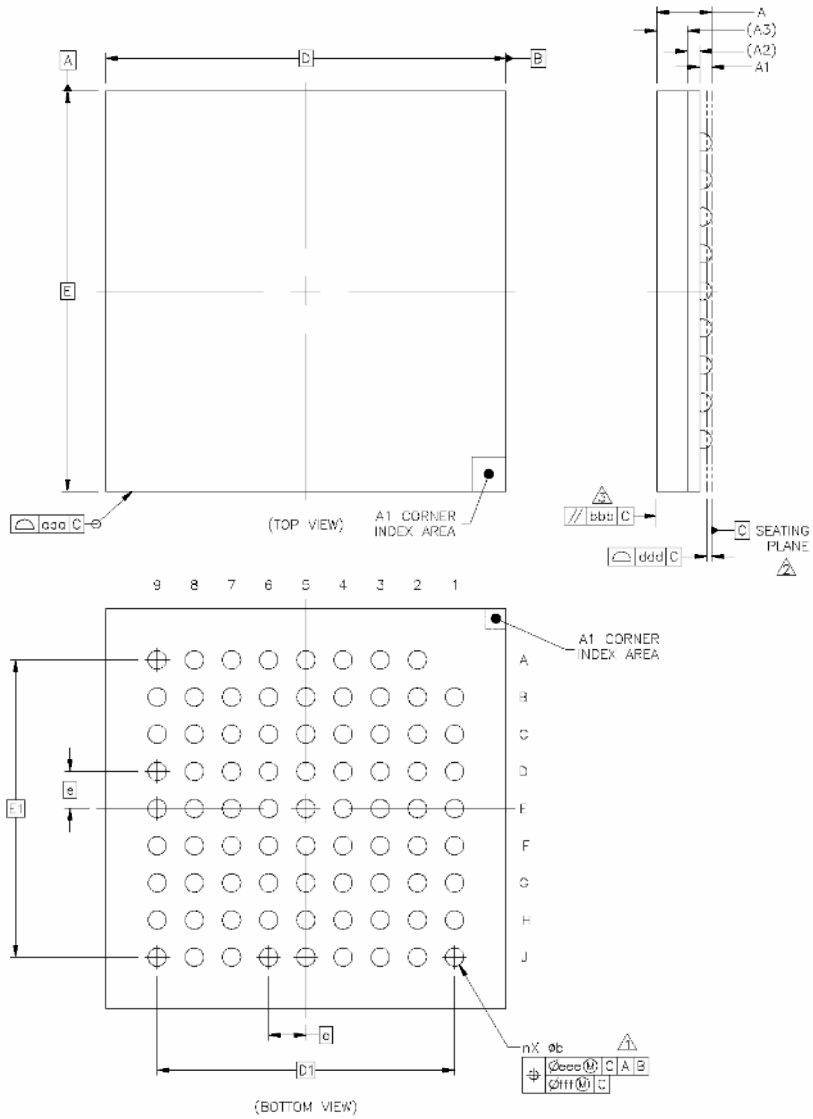


Figure 14-2 : Application circuit for SSD1963 (With crystal oscillator input)



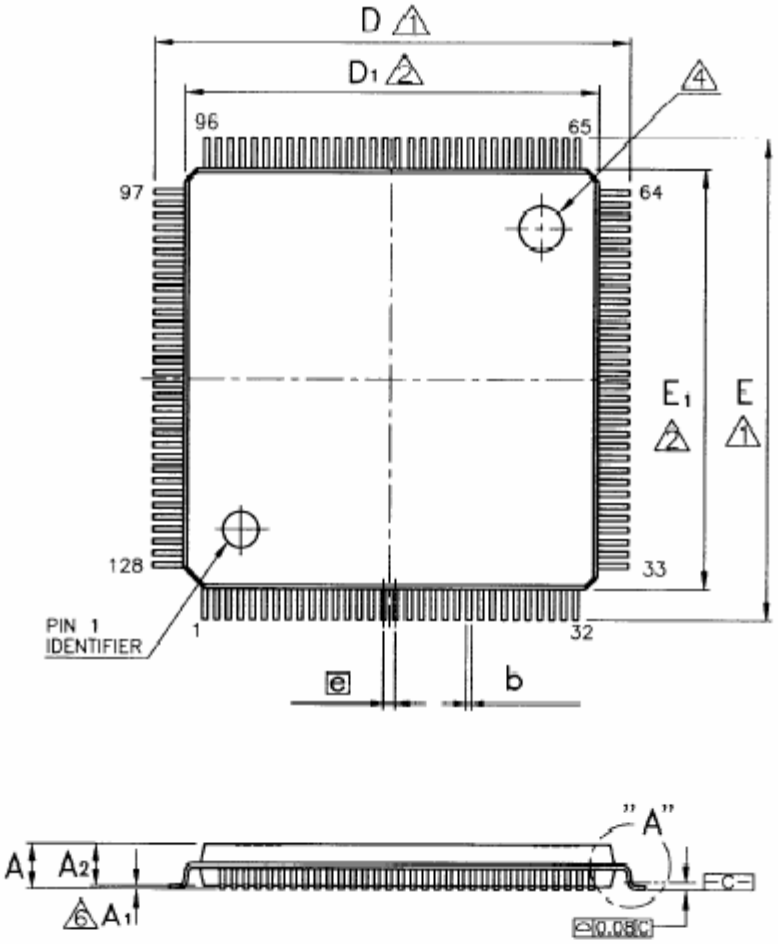
15 PACKAGE INFORMATION

15.1 Package Mechanical Drawing for 80 balls TFBGA



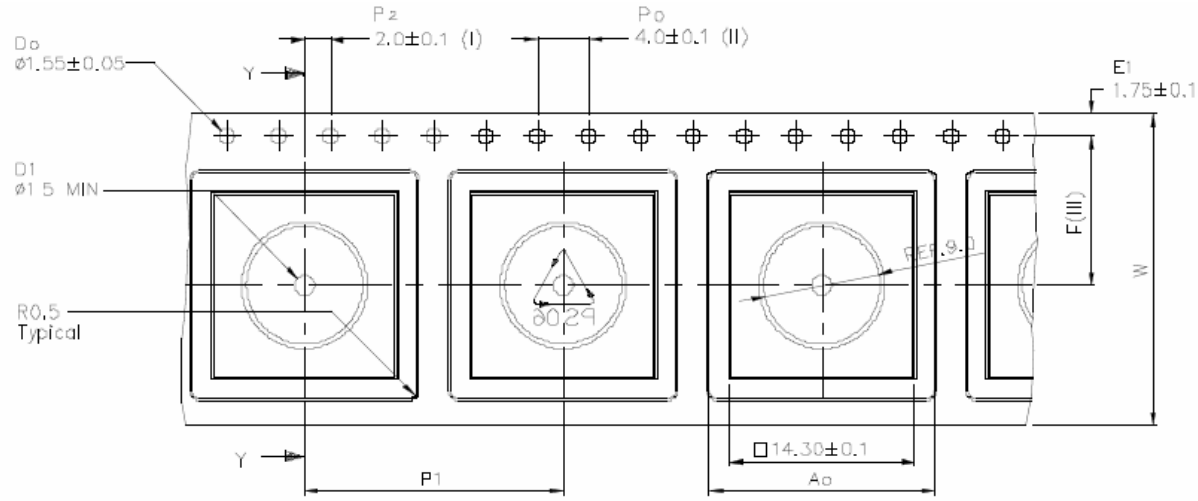
Dimension in mm			
Symbol	**Min**	**Typical**	**Max**
A	--	---	1.1
A1	0.16	---	0.26
A2	---	0.21	---
A3	---	0.54	---
b	0.27	---	0.37
D	---	7.00 BSC	---
E	---	7.00 BSC	---
e	---	0.65 BSC	---
D1	---	5.2 BSC	---
E1	---	5.2 BSC	---
aaa	---	0.1	---
bbb	---	0.2	---
ddd	---	0.08	---
eee	---	0.15	---
fff	---	0.08	---

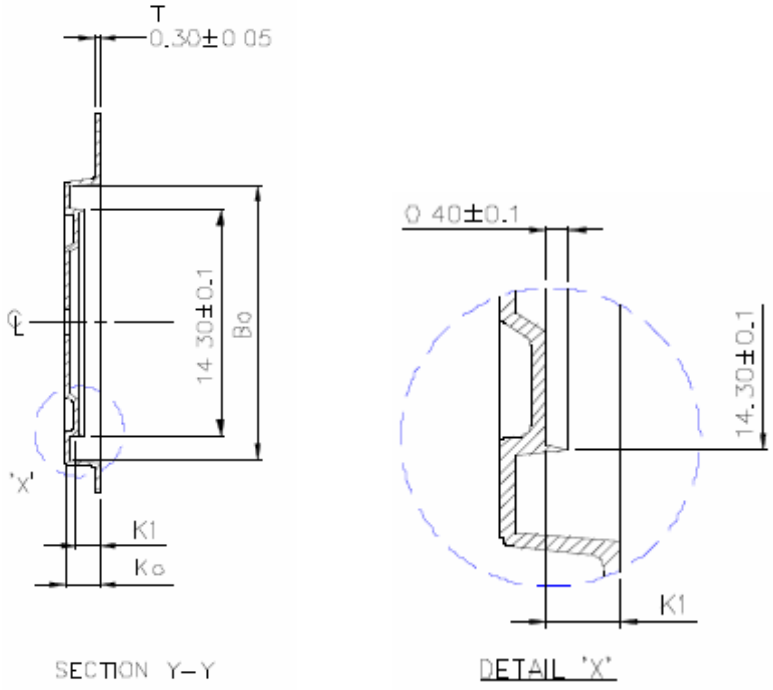
15.2 Package Mechanical Drawing for 128 pins LQFP



Symbol	Dimension in mm		
	Min	Nom	Max
A			1.60
A1	0.05		
A2		1.40	
D		16.00	
D1		14.00	
E		16.00	
E1		14.00	
e	0.40 BSC		
b		0.18	

15.3 Tape & Reel Drawing for 128 pins LQFP





Ao	17.40	+/-	0.1
Bo	17.40	+/-	0.1
Ko	2.00	+/-	0.1
K1	1.40	+/-	0.1
F	11.50	+/-	0.1
P1	20.00	+/-	0.1
W	24.00	+/-	0.3

(I) Measured from centreline of sprocket hole to centreline of pocket

(II) Cumulative tolerance of 10 sprocket holes is ± 0.20 .

(III) Measured from centreline of sprocket hole to centreline of pocket.

(IV) Ao and Bo measured on a plane 0.3mm above the btm of the pocket.

(V) Other material available

ALL DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED.

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